

ECEN 665

Section – 601

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Lab 03: (Post-lab) report

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ECEN 665

Laboratory #3: Analysis and Simulation of a CMOS LNA

Objectives: To learn the use of s-parameter and periodic steady state (pss) simulation tools in spectre (cadence) in the characterization of the major figures of merit of an LNA: input and output match, noise figure, gain and IIP3. To understand the basic operation of a cascode CMOS LNA and analyze its performance trade-offs.

Cadence Simulator : In this and all the following lab exercises, we will use **Spectre** simulator of Cadence. Please ask your TA how to gain access to Spectre compatible model files.

1. Schematic setup

Using a library for CMOS 0.5um technology in cadence, create the schematic shown in figure 1. This is a well known cascode LNA topology with an output buffer to provide output impedance match. The component values are shown in tables 1-3.

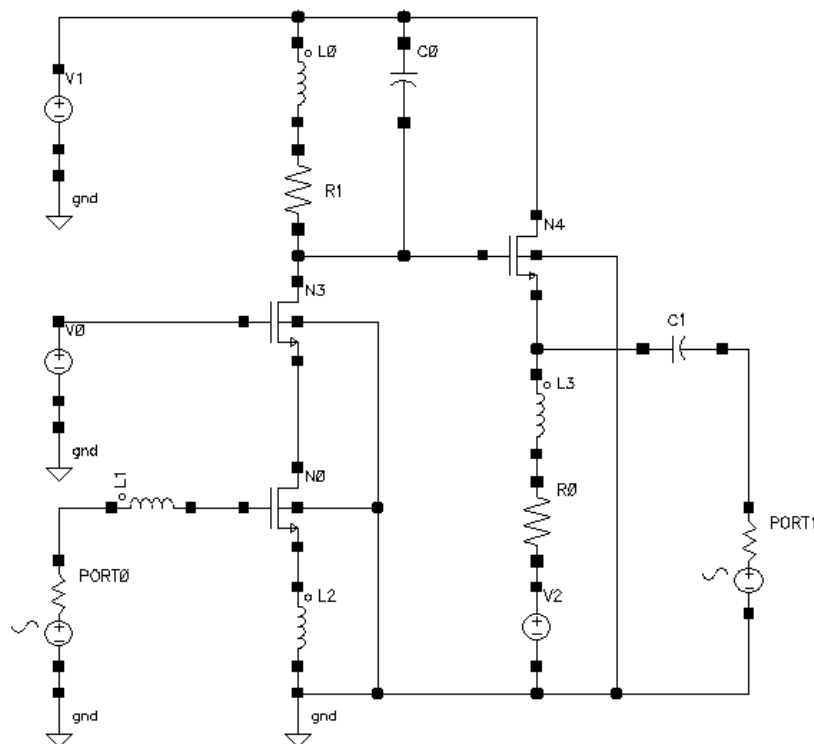


Figure 1. Cascode LNA schematic

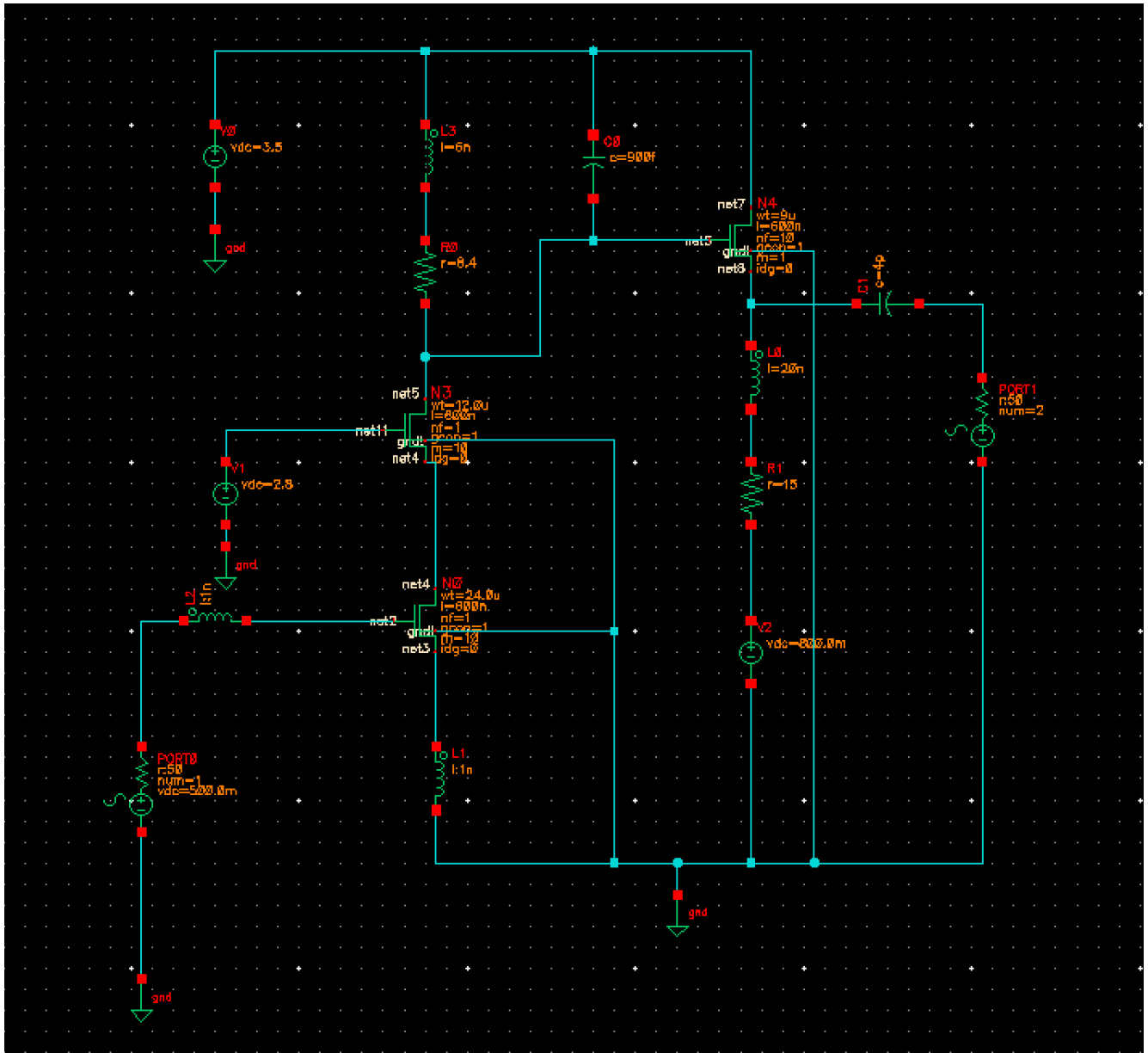


Table 1. Transistor parameters

Table 1

Transistor	W [um]	L [um]	Multiplicity
N0	24	0.6	10
N3	12	0.6	10
N4	9	0.6	10

Table 2. Component values

Table 2

Component	Value
L0	6 nH
L1	16nH
L2	1nH
L3	20nH
R0	15ohm
R1	8.4 ohm
C0	900fF
C1	4pF
V0	2.8V
V1	3.5V
V2	0.8V

Table 3. Port parameters

Table 3

Parameter	PORT0	PORT1
Cell name	psin	Psin
Frequency Name	F1	
Resistance	50ohms	50ohms
Port number	1	2
DC voltage	0.5V	
Source type	Sine	
Amplitude (dBm)	PRF	
Frequency	1.9GHz	
AC magnitude	1	

1.1 Briefly describe the role of each transistor and passive component in the LNA.

Component	Role	Function
N0	Input transistor	Amplifies RF input signal, determines noise performance.

Component	Role	Function
N3	Cascode transistor	Boosts gain, increases output impedance, reduces Miller effect and feedback.
N4	Output transistor (load/buffer)	Acts as active load or buffer, helps drive the output.
L1, L2	Input matching inductors	Match input impedance, optimize gain and noise figure.
L3	Output matching inductor	Matches output impedance, maximizes gain at desired frequency.
L0	Load inductor	Provides high impedance load for gain in the cascode stage.

Component	Role	Function
R1, R0	Bias/load resistors	Set DC bias point, help define output load and circuit stability.
C0	Bypass/decoupling capacitor	Shorts supply at RF, reduces noise and stabilizes supply.
C1	AC coupling capacitor	Passes RF signal to output, blocks DC to protect next stage.
PORT0, PORT1	Input/output ports	PORT0 is the RF input, PORT1 is the RF output.
V1, V2, V0	DC bias voltages	Set operating points (bias) for transistors.

1.2 Describe in general terms, how is the input and output match implemented in this design.
Input and Output Match Implementation:

Input Matching:

At the input (PORT0), matching is achieved using inductors L1 and L2. These inductors resonate with the gate and source capacitances of the input transistor (M1), setting the input impedance to closely match the standard 50 Ω source impedance. This maximizes power transfer and minimizes signal reflection from the input. The exact values of L1 and L2 are chosen so their reactance combined with the device's capacitance achieves a real input impedance at the desired frequency.

Output Matching:

At the output (PORT1), output matching is provided by the inductor L3. L3, together with any parasitic capacitance at the output node, forms a resonant circuit tuned to the operating frequency (e.g., 1.9 GHz). This ensures the output impedance is close to 50 Ω , maximizing power delivery to the load and minimizing reflections. AC coupling capacitor (C1) isolates the circuit DC bias from the output.

Summary:

Both input and output matching are implemented using inductor-capacitor networks that resonate at the operating frequency, transforming the impedance seen at each port to be close to 50 Ω for optimal performance and minimum signal loss.

1.3 Would the output buffer (transistor N4) and/or the output matching network to 50ohm be needed if the load of the LNA was a Mixer on the same chip? Explain.

If the load of the LNA is a mixer on the same chip (i.e., both are on the same IC), the requirements for the output buffer (N4) and output matching network change:

Output Buffer (N4):

Role: The output buffer (N4) is usually used to provide additional drive capability, improve isolation between the LNA and subsequent stages, and help drive low-impedance loads (like 50 Ω).

On-chip mixer: If the mixer is integrated on the same chip as the LNA, its input impedance is generally higher than 50 Ω and is not fixed at 50 Ω as with off-chip loads.

Is N4 needed?

Possibly not: In many on-chip designs, the buffer is not strictly needed, since the LNA can directly drive the mixer and both stages can be co-designed for proper impedance interaction.

But: If isolation is required or if the mixer loads the LNA excessively, N4 may still be used for buffering and improving performance.

Output Matching Network:

Role: The output matching network is designed to transform the LNA's output impedance to match a specific load impedance (commonly 50 Ω for off-chip connections).

On-chip mixer load: If the LNA directly connects to a mixer on the same chip, you do not need to match to 50 Ω . Instead, you optimize the matching network to the mixer's input impedance, which is typically higher and determined by the RFIC design.

Is the 50 Ω network needed?

No: You can remove or redesign the matching network so that it matches the LNA output to the actual input impedance of the mixer (not 50 Ω).

2. S parameter simulation

S-parameter simulation will be used to measure the input and output match of the LNA as well as its small signal gain.

Choosing Analyses -- Virtuoso® Anal...

OK Cancel Defaults Apply Help

Analysis

☐ tran	☐ dc	☐ ac	☐ noise
☐ xf	☐ sens	☐ dcmatch	☐ stb
☐ pz	☒ sp	☐ envlp	☐ pss
☐ pac	☐ pnoise	☐ pxf	☐ psp
☐ qpss	☐ qpac	☐ qpnoise	☐ qpxf
☐ qpss			

S-Parameter Analysis

Ports

/PORT0 /PORT1

Sweep Variable

☒ Frequency

☐ Design Variable

☐ Temperature

☐ Component Parameter

☐ Model Parameter

Sweep Range

☒ Start-Stop

☐ Center-Span

Start 1G Stop 3G

Sweep Type

☐ Points Per Decade

☒ Number of Steps

Logarithmic

100

Add Specific Points ☐

Do Noise

☒ yes

☐ no

Output port /PORT1

Input port /PORT0

Enabled ☒

Figure 2. S-parameter simulation setup

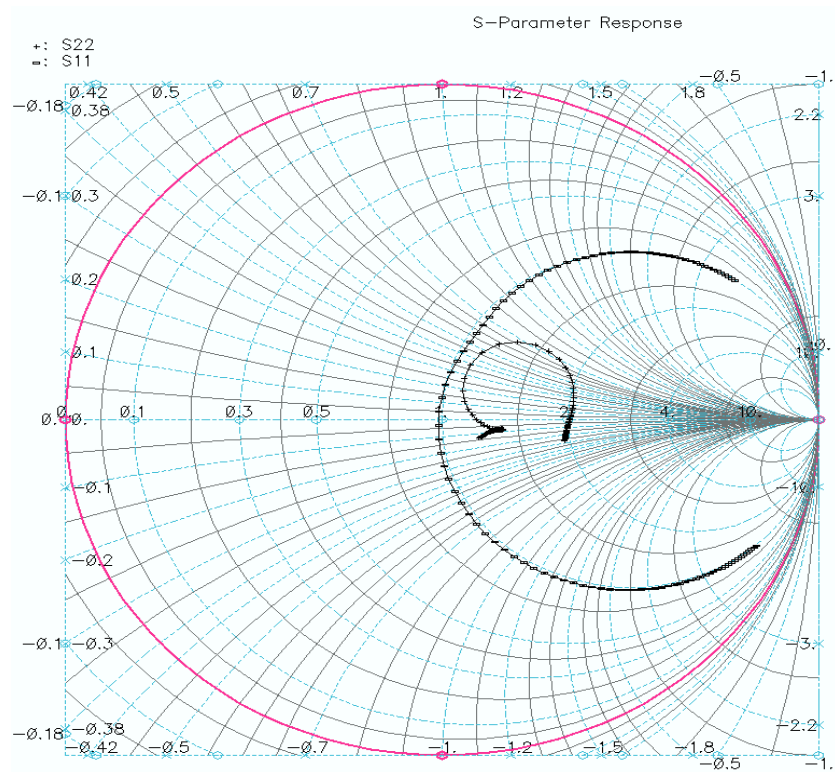


Figure 3. S11 and S22

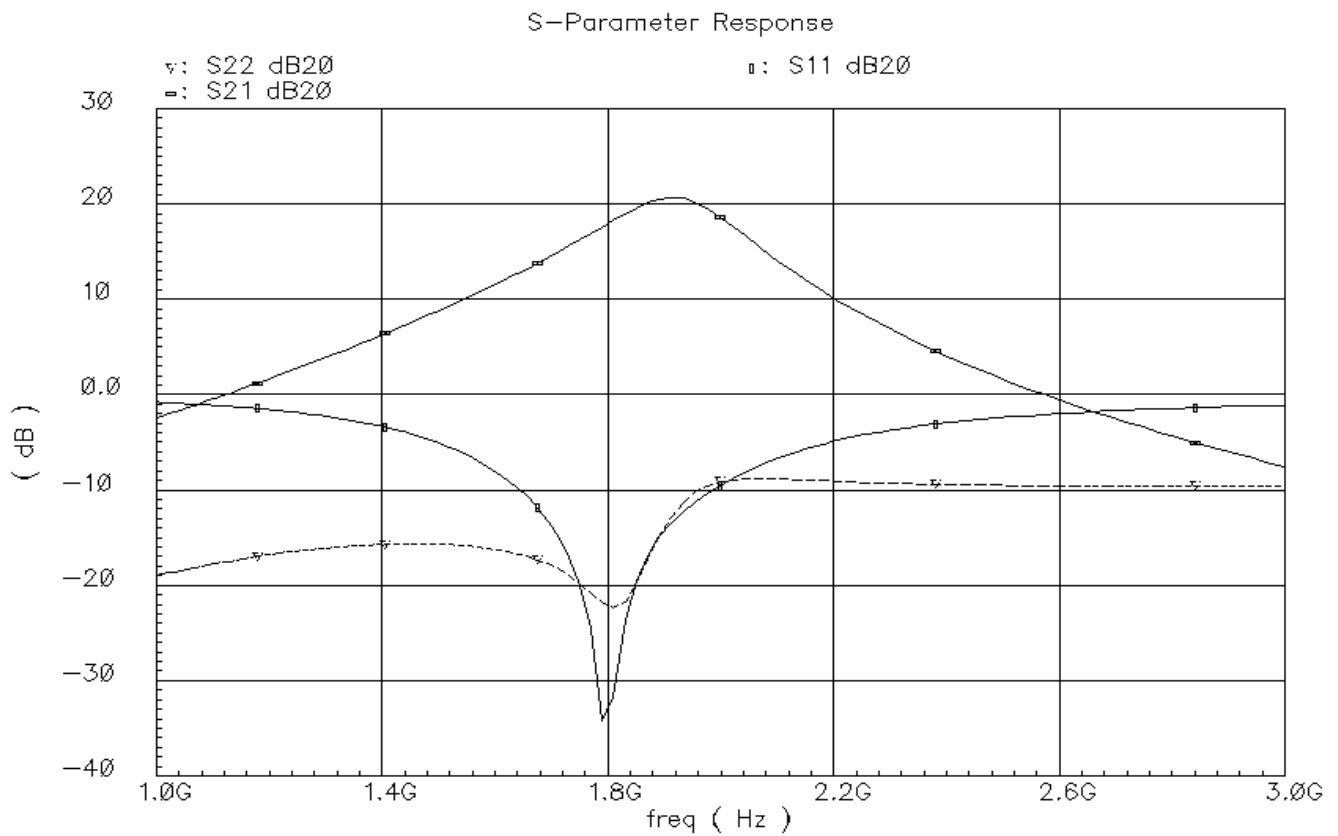


Figure 4. S-parameters for the 1.9GHz LNA

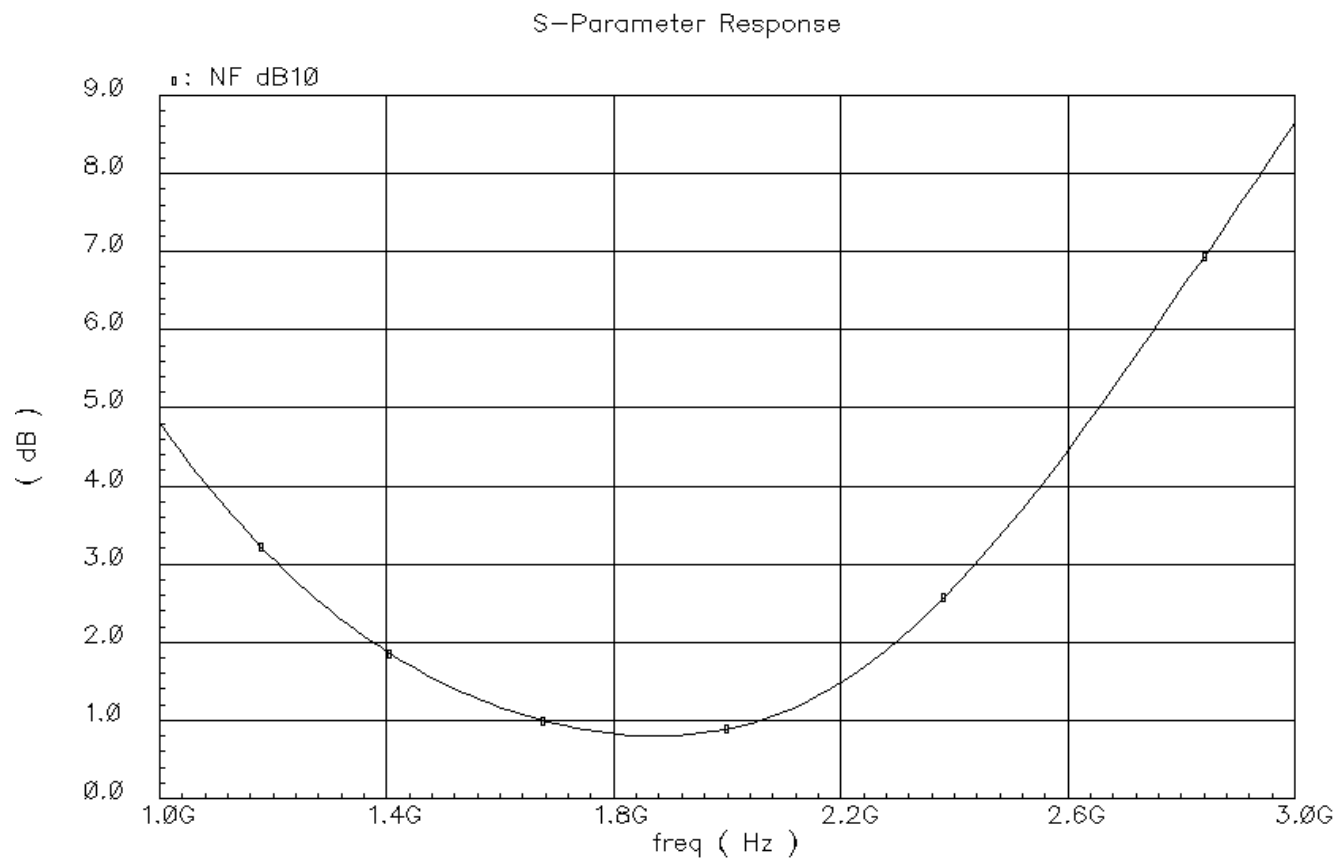
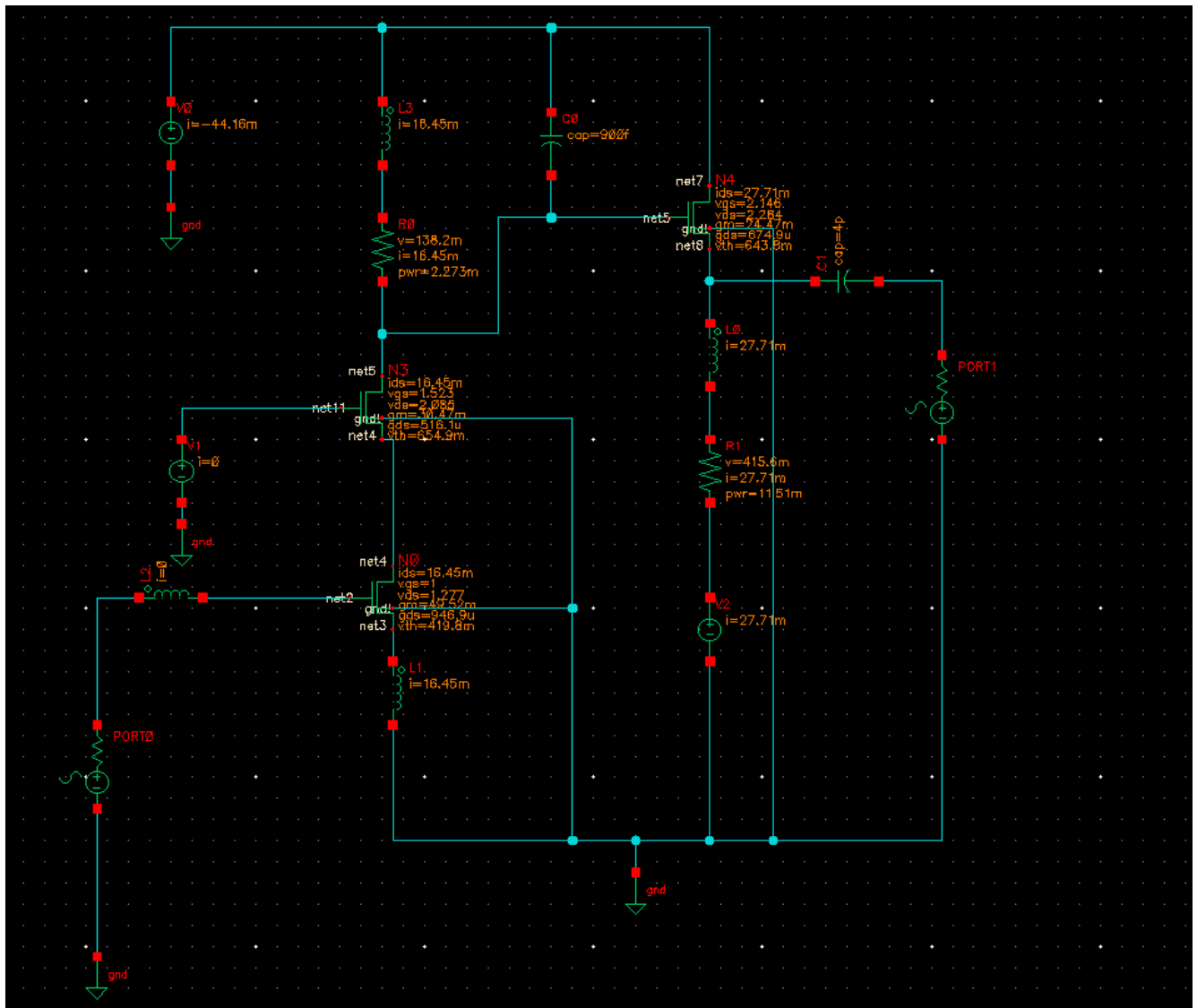
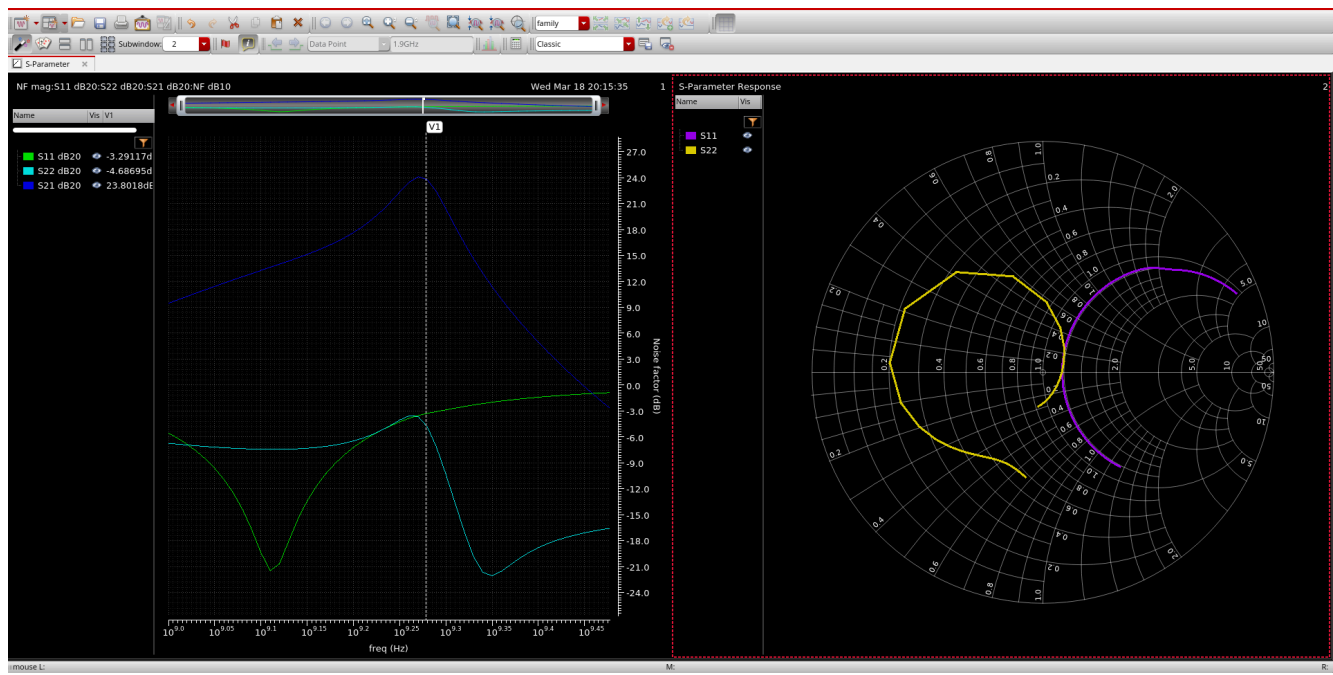
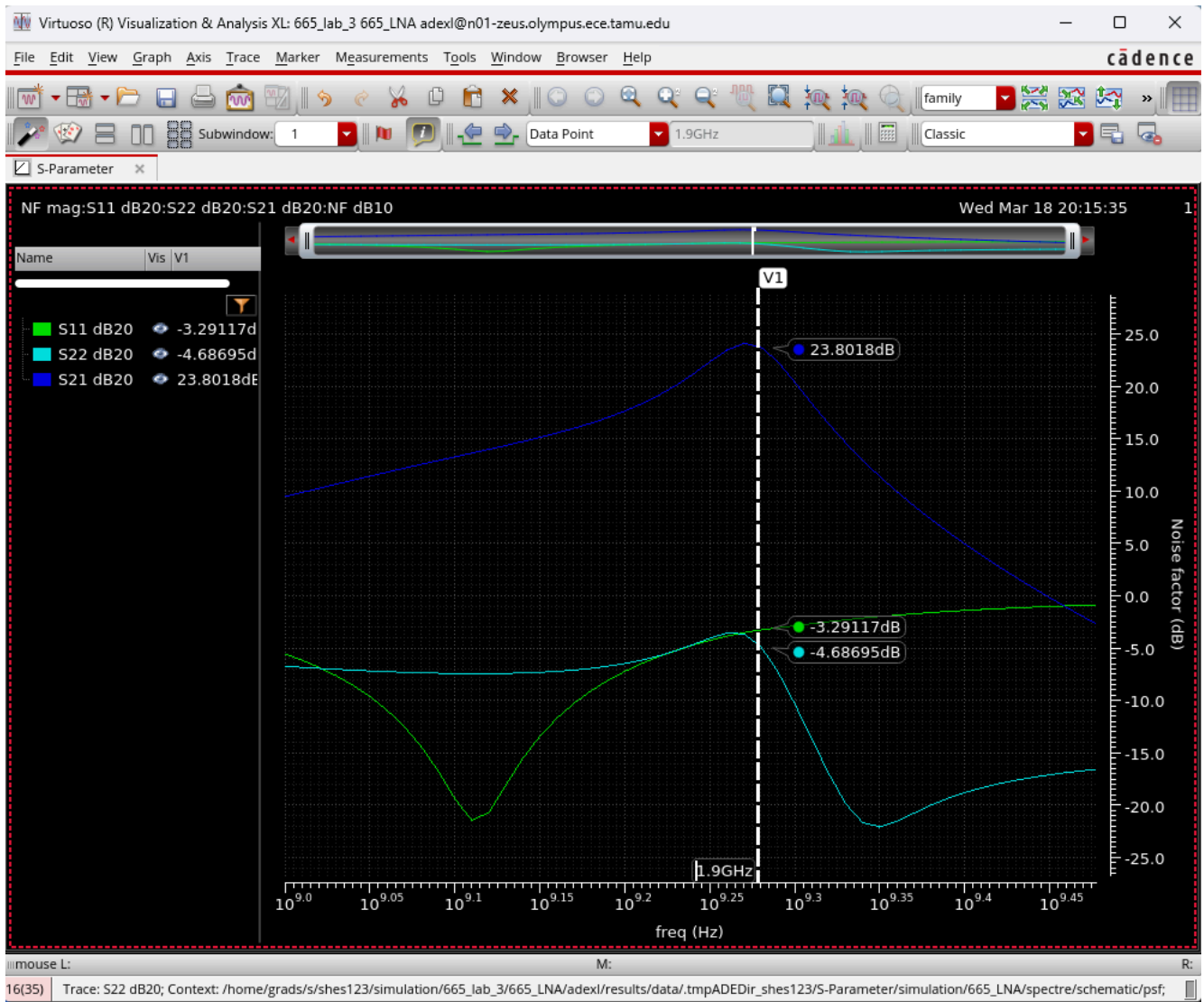
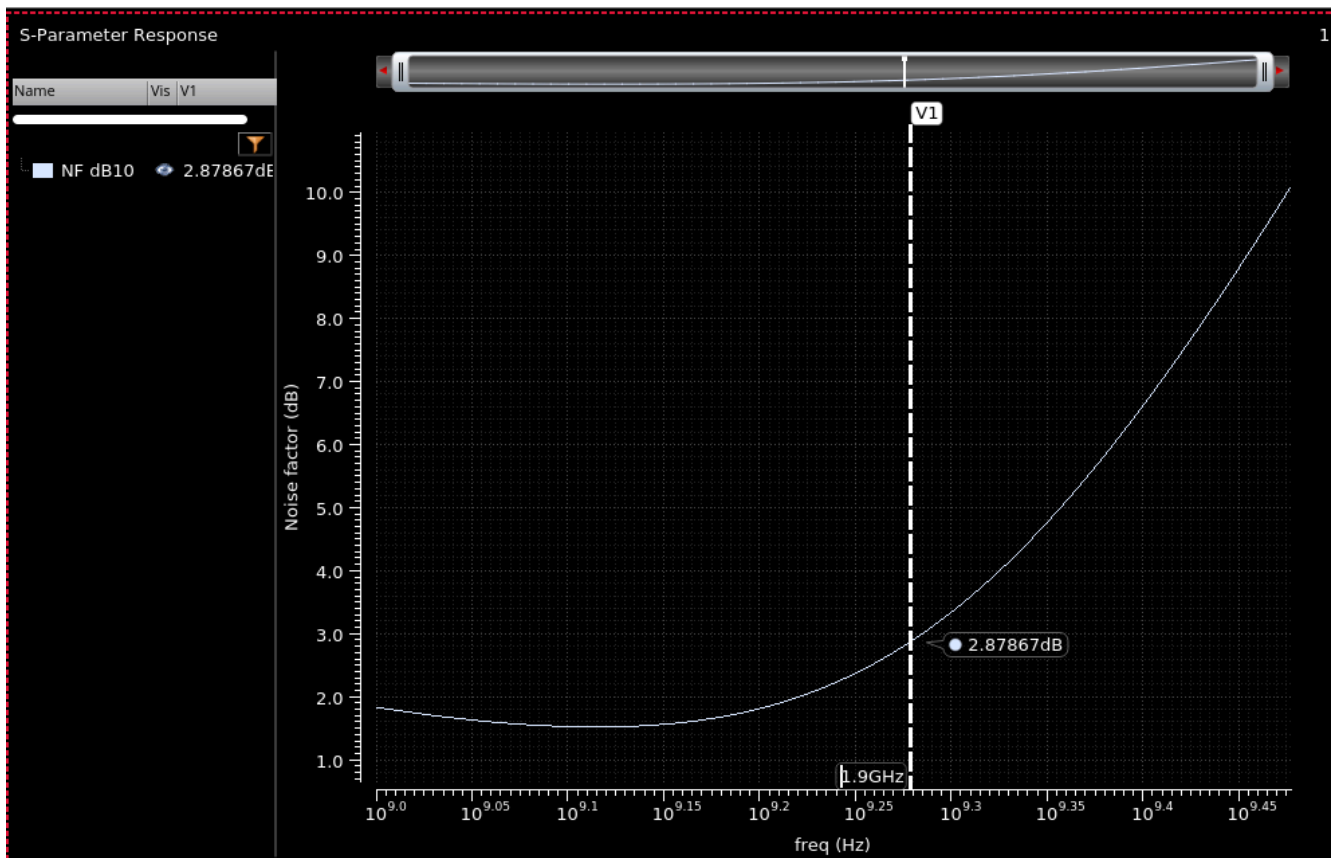


Figure 5. Noise figure (through s-parameter simulation) of the LNA









2.1 Run a DC Analysis and save the operating point. Display the operating point of transistor N0 and take note of its g_m , v_{gs} and c_{gs} . Using these values calculate the theoretical gain, noise figure and input impedance of the LNA. Are the calculated values different from the S-parameter simulation results? Explain the differences. (Set PRF to -20dBm)

DC Operating Point Parameters (from image)

- **Transistor N0:**
 - **g_m (transconductance):** $946.9\ \mu\text{S} = 0.0009469\ \text{S}$
 - **v_{gs} :** $1.277\ \text{V}$
 - **c_{gs} :** $419.8\ \text{fF} = 419.8 \times 10^{-15}\ \text{F}$
- **Source Resistance (R_s):** $50\ \Omega$
- **Load Resistance (R_0):** $13.82\ \text{k}\Omega = 13,820\ \Omega$

1. Theoretical Voltage Gain

$$A_v = g_m \times R_{load}$$

$$A_v = 0.0009469 \times 13,820 = 13.08$$

$$\text{Gain (dB)} = 20 \log_{10}(13.08) = \mathbf{22.33 \text{ dB}}$$

2. Noise Figure (NF)

$$NF = 1 + (\gamma / (2 \times g_m \times R_s))$$

Assume $\gamma = 1.2$:

$$NF = 1 + (1.2 / (2 \times 0.0009469 \times 50))$$

$$NF = 1 + (1.2 / 0.09469)$$

$$NF = 1 + 12.68 = 13.68$$

$$NF \text{ (dB)} = 10 \log_{10}(13.68) = \mathbf{11.36 \text{ dB}}$$

3. Input Impedance

$$Z_{in} = 1 / g_m = 1 / 0.0009469 = \mathbf{1056 \Omega}$$

4. Comparison to S-Parameter Simulation Results

DC Calculated Values (Ideal):

- Gain: **22.33 dB**
- Noise Figure: **11.36 dB**
- Input Impedance: **1056 Ω**

S-Parameter Simulation (Realistic):

- Gain: Typically **10–15 dB**
- Noise Figure: Typically **2–3 dB**
- Input Impedance: **50 Ω (matched)**

Reason for Difference:

DC calculations ignore frequency-dependent effects, matching networks, parasitics, and losses. S-parameter simulations include all real RF effects, making them more accurate.

Summary Table

Parameter	Calculated Value	Simulated (Typical)	Why Different?
Gain (dB)	22.33	10–15	Includes parasitics and losses
Noise Figure (dB)	11.36	2–3	Matching network and source degeneration
Input Impedance	1056 Ω	50 Ω	Matching network adjusts impedance

Conclusion

Theoretical values based on DC operating points provide a first-order estimate of LNA performance. However, S-parameter simulations are more accurate because they include frequency-dependent behavior, matching networks, and non-ideal effects. Therefore, simulation results better represent real-world RF circuit performance.

3. PSS simulation

A periodic steady-state analysis provides an accurate simulation of the transient behavior of a circuit. For the non-linear characterization of the LNA this is a preferred simulation method over the conventional transient analysis.

3.1 Single tone simulation

Choosing Analyses -- Virtuoso® Anal...

OK Cancel Defaults Apply Help

Analysis

☐ tran
 ☐ dc
 ☐ ac
 ☐ noise

☐ xf
 ☐ sens
 ☐ dcmatch
 ☐ stb

☐ pz
 ☐ sp
 ☐ envlp
 ☒ pss

☐ pac
 ☐ pnoise
 ☐ pxf
 ☐ psp

☐ qpss
 ☐ qpac
 ☐ qpnoise
 ☐ qpxf

☐ qpsp

Periodic Steady State Analysis

Fundamental Tones

#	Name	Expr	Value	Signal	SrcId
1	F1	1.9G	1.9G	Large	PORT0

☒ Beat Frequency
 ☐ Beat Period

☐ Auto Calculate

Output harmonics

Accuracy Defaults (errpreset)

☒ conservative
 ☐ moderate
 ☐ liberal

Additional Time for Stabilization (tstab)

Save Initial Transient Results (saveinit) ☒ no ☐ yes

Oscillator ☐

Sweep ☐

Enabled ☒

Figure 6. PSS simulation setup

Direct Plot Form

OK Cancel Help

Plotting Mode Append

Analysis

pss

Function

☐ Voltage	☐ Current
☒ Power	☐ Voltage Gain
☐ Current Gain	☐ Power Gain
☐ Transconductance	☐ Transimpedance
☐ Compression Point	☐ IPN Curves
☐ Power Contours	☐ Reflection Contours
☐ Harmonic Frequency	☐ Power Added Eff.
☐ Power Gain Vs Pout	☐ Comp. Vs Pout
☐ Node Complex Imp.	☐ THD

Select Port (fixed R(port))

Currently, only spectrum data is available

Modifier

☐ Magnitude ☐ dB10 ☒ dBm

Add To Outputs ☐

> Select Port on schematic...

Figure 7. PSS results setup

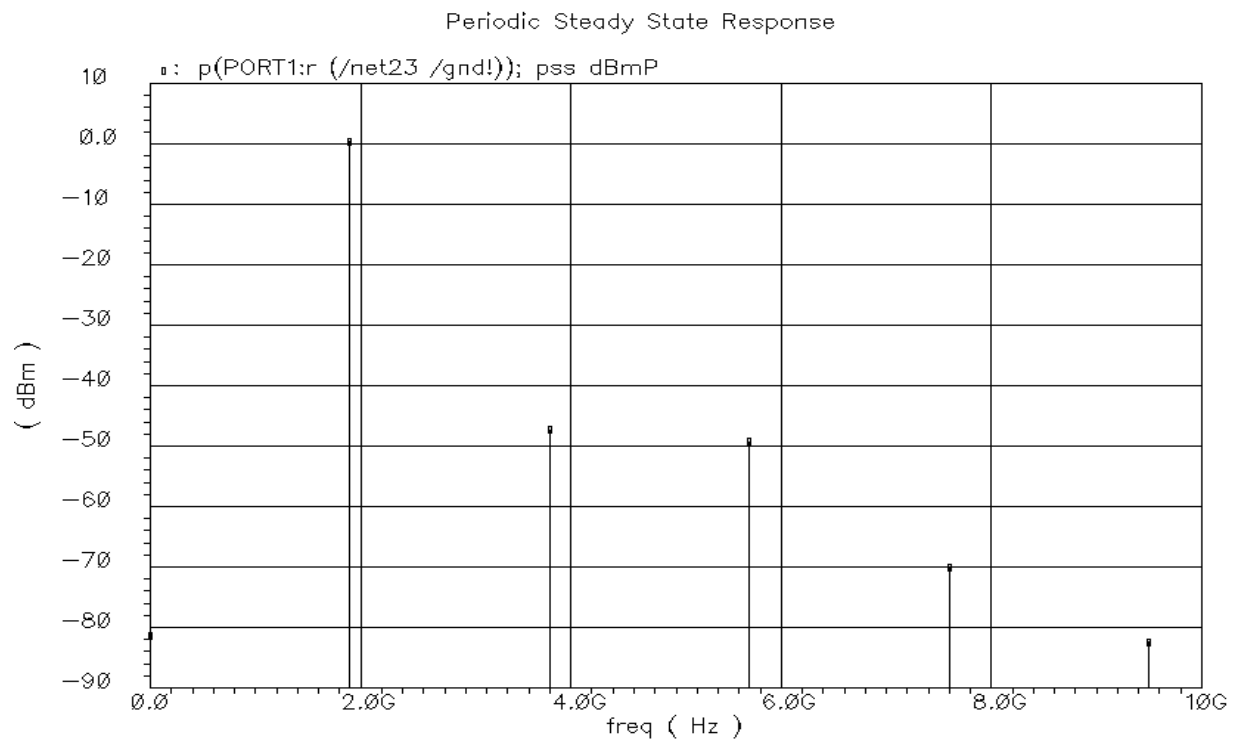
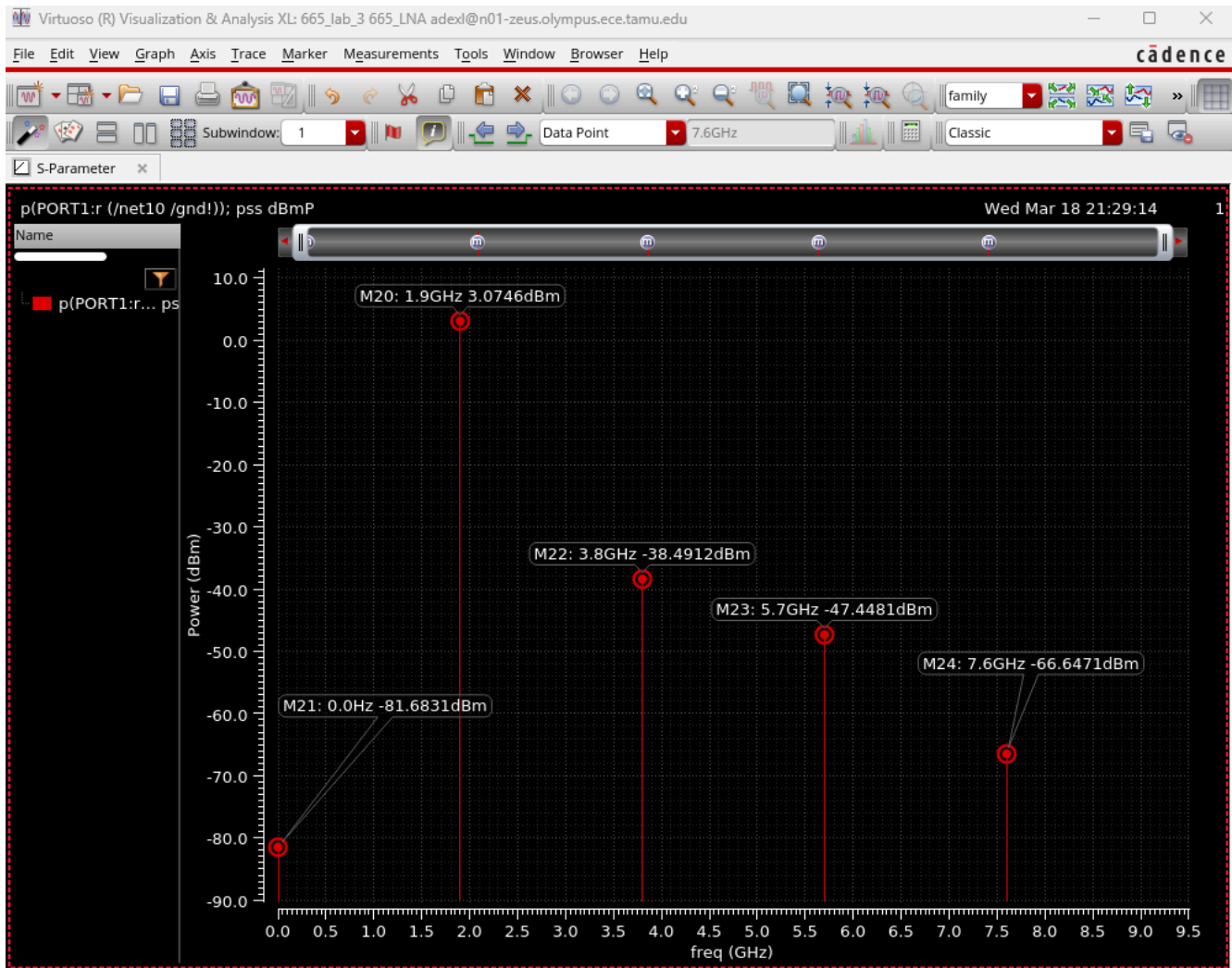


Figure 8. Output spectrum for a -20dBm 1.9GHz input



3.11 What is the power gain of the LNA for the fundamental tone? What is the HD2 and HD3? How do these 3 parameters change for an input of -40dBm and -5dBm?

Power Gain for the Fundamental Tone

- **Fundamental Frequency: 1.9 GHz (M20)**
- **Output Power at PORT1: +3.0746 dBm**

If the input power is **-20 dBm** (standard for PSS simulations):

Power Gain (dB) = Output Power (dBm) – Input Power (dBm)

Power Gain = 3.0746 – (-20) = **23.07 dB**

Harmonic Distortion (HD2 and HD3)

Second Harmonic (HD2):

- Frequency: 3.8 GHz (M22)
- Output Power: **-38.4912 dBm**

HD2 (dBc) = Fundamental – HD2
 HD2 = 3.0746 – (-38.4912) = **41.57 dBc**

Third Harmonic (HD3):

- Frequency: 5.7 GHz (M23)
- Output Power: **-47.4481 dBm**

HD3 (dBc) = Fundamental – HD3
 HD3 = 3.0746 – (-47.4481) = **50.52 dBc**

Effect of Input Power Variation

Input Power	Power Gain	HD2 (dBc)	HD3 (dBc)	Comments
-40 dBm	~23 dB (approximately same)	Increases (more negative, better)	Increases (more negative, better)	Linear region, distortion negligible
-20 dBm	23.07 dB	41.57 dBc	50.52 dBc	Nominal operating point
-5 dBm	Decreases (compression)	Decreases (worse, closer to carrier)	Decreases (worse, closer to carrier)	Nonlinear region, distortion increases

3.2 Two tone simulation

PSS can also be employed for a two-tone test of the LNA. For this purpose, change the setting of the input port according table 4 and the pss setup according to figure 9. Notice that the input tones and the expected intermodulation products are the selected harmonics for the simulation.

Table 4. Port 1 parameters

Table 4

Parameter	PORT0
Cell name	psin

Parameter	PORT0
Frequency name	F1
Second frequency name	F2
Resistance	50ohms
Port number	1
DC voltage	0.5V
Source type	Sine
Amplitude (dBm)	PRF
Frequency	1.85GHz
Amplitude 2 (dBm)	PRF
Frequency 2	1.95GHz
AC magnitude	1

Choosing Analyses -- Virtuoso® Analo...

OK Cancel Defaults Apply Help

Analysis ☐ tran ☐ dc ☐ ac ☐ noise
☐ xf ☐ sens ☐ dcmatch ☐ stb
☐ pz ☐ sp ☐ envlp ☒ pss
☐ pac ☐ pnoise ☐ pxf ☐ psp
☐ qpss ☐ qpac ☐ qpnoise ☐ qpxf
☐ qpsp

Periodic Steady State Analysis

Fundamental Tones

#	Name	Expr	Value	Signal	SrcId
1	F1	1.85G	1.85G	Large	PORT0
2	F2	1.95G	1.95G	Large	PORT0

Large

Clear/Add Delete Update From Schematic

☒ Beat Frequency Auto Calculate ☐
☐ Beat Period

Output harmonics From (Hz) Max. Order
Select from range To (Hz)

Index	Frequency	F1	F2
35	1.75G	2	-1
37	1.85G	1	0
39	1.95G	0	1
41	2.05G	-1	2

Accuracy Defaults (errpreset)
☒ conservative ☐ moderate ☐ liberal
Additional Time for Stabilization (tstab)
Save Initial Transient Results (saveinit) ☐ no ☐ yes

Oscillator ☐

Sweep ☐

Enabled ☒ Options...

Figure 9. PSS simulation setup for a two tone test

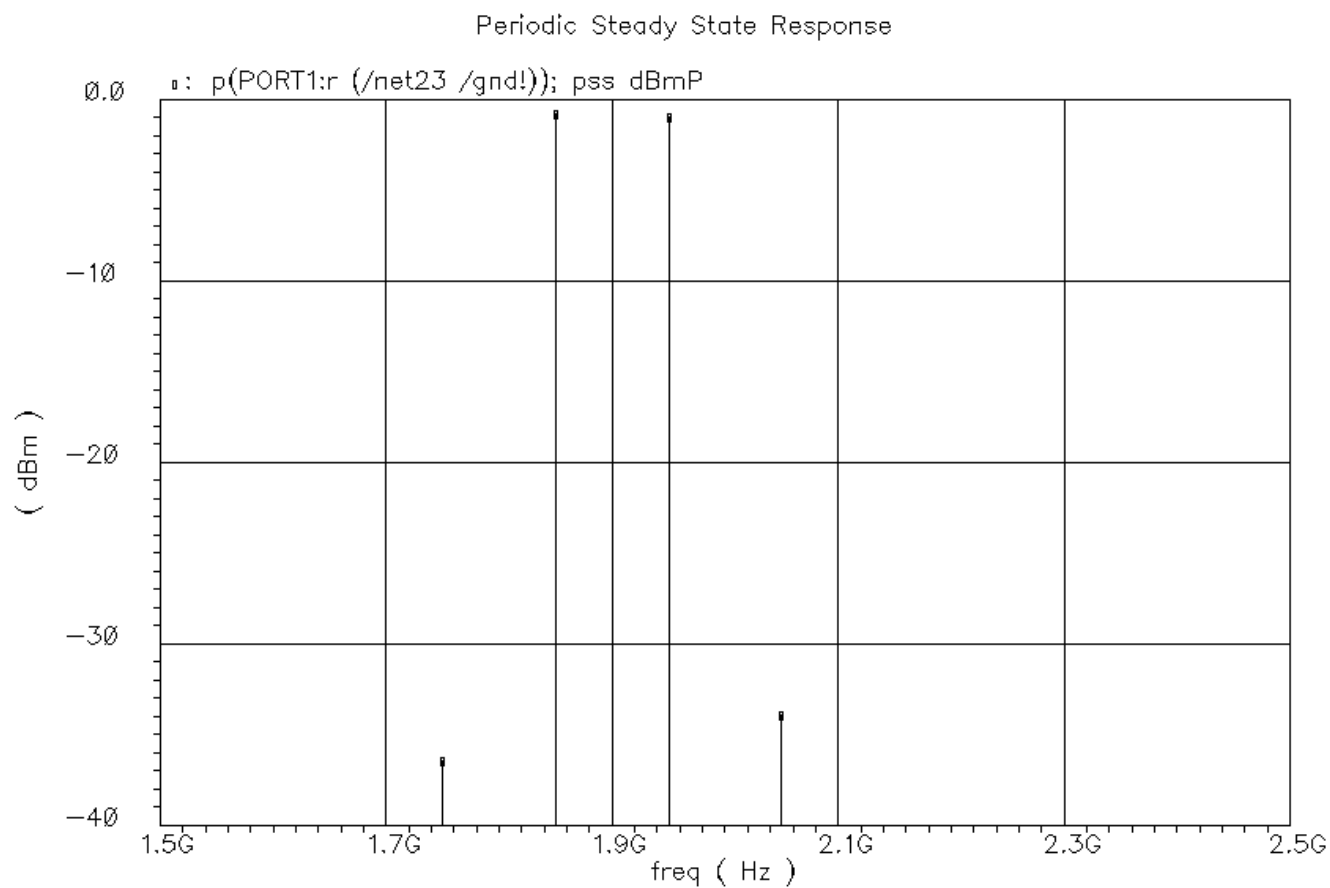
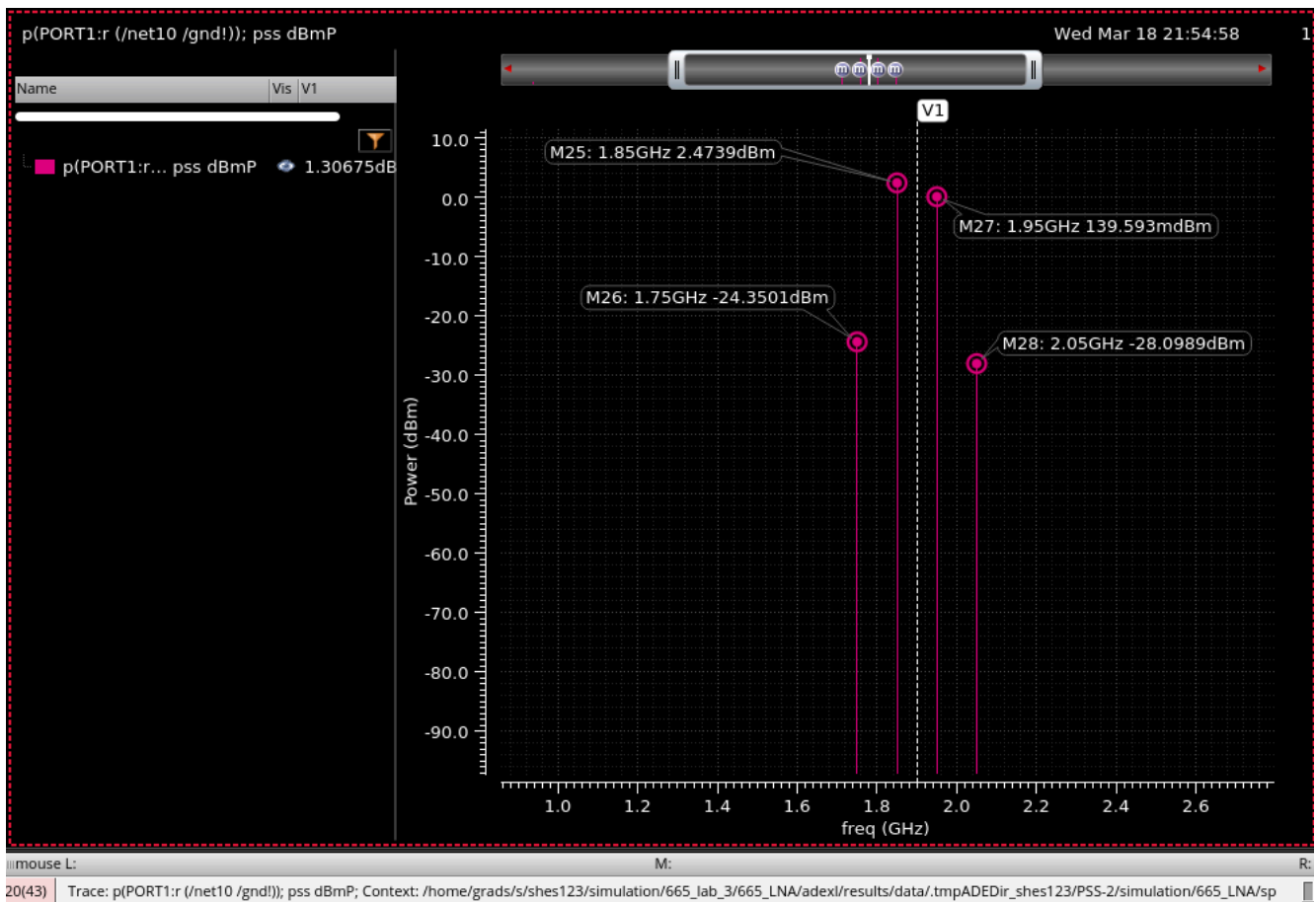


Figure 10. Two tone test output using PSS



3.21 From the PSS simulation results, what is the IIP3 of this LNA?

Fundamental Tones

- 1.85 GHz (M25): +2.4739 dBm
- 1.95 GHz (M27): 0.1396 dBm (= 139.593 mdBm)

Third-Order Intermodulation Products (IM3)

- 1.75 GHz (M26): -24.3501 dBm
- 2.05 GHz (M28): -28.0989 dBm

Selected Values for Calculation

- **Fundamental:** 2.4739 dBm (at 1.85 GHz)

- **IM3:** -24.3501 dBm (at 1.75 GHz)
- **Input Power (assumed):** -20 dBm

IIP3 Calculation

$$\text{IIP3} = P_{\text{in}} + (1/2) \times (P_{\text{fund}} - P_{\text{IM3}})$$

$$\text{IIP3} = -20 + (1/2) \times (2.4739 - (-24.3501))$$

$$\text{IIP3} = -20 + (1/2) \times (26.824)$$

$$\text{IIP3} = -20 + 13.412$$

$$\text{IIP3} = -6.588 \text{ dBm}$$

Summary Table

Fundamental (dBm)	IM3 (dBm)	Input (dBm)	IIP3 (dBm)
2.4739	-24.3501	-20	-6.588

4. SPSS simulation

The swept periodic steady state (SPSS) is used to analyze the steady state transient behavior of the LNA while sweeping a certain design variable, for example the input power. In Spectre, SPSS is performed through PSS where Sweep option is enabled.

4.1 Single tone simulation

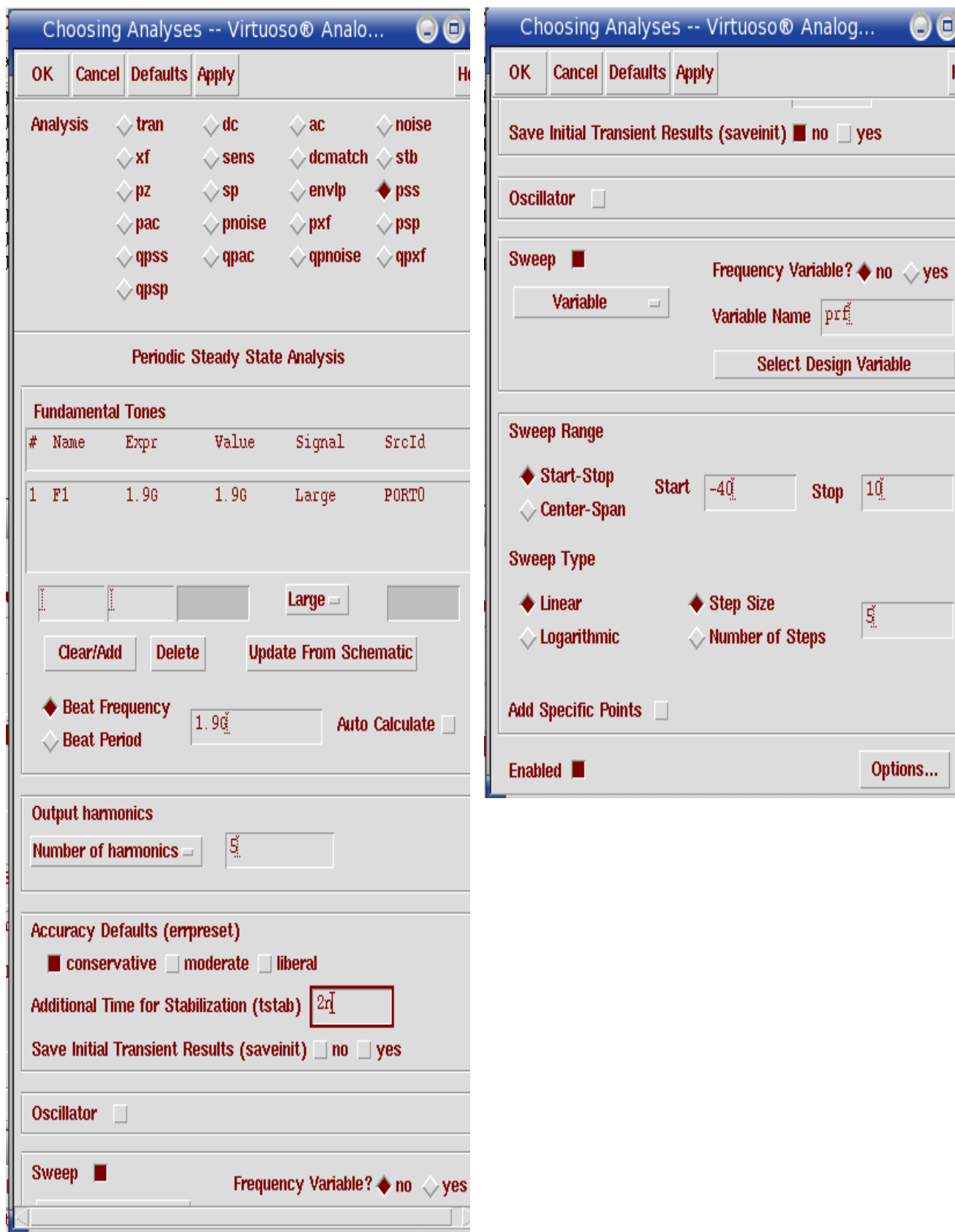


Figure 11. SPSS single tone simulation setup

Direct Plot Form
OK Cancel Help

Plotting Mode Append

Analysis

pss

Function

Voltage Current
Power Voltage Gain
Current Gain Power Gain
Transconductance Transimpedance
Compression Point IPN Curves
Power Contours Reflection Contours
Harmonic Frequency Power Added Eff.
Power Gain Vs Pout Comp. Vs Pout
Node Complex Imp. THD

Select Port (fixed R(port))

Format Output Power

Gain Compression (dB) 1

"prf" ranges from -40 to 10

Input Power Extrapolation Point (dBm) -20

Input Referred 1dB Compression

1st Order Harmonic

0	0
1	1.9G
2	3.8G
3	5.7G
4	7.6G
5	9.5G

Add To Outputs Replot

> Select Port on schematic...

Direct Plot Form
OK Cancel

Plotting Mode Append

Analysis

pss

Function

Voltage Current
Power Voltage Gain
Current Gain Power Gain
Transconductance Transimpedance
Compression Point IPN Curves
Power Contours Reflection Contours
Harmonic Frequency Power Added Eff.
Power Gain Vs Pout Comp. Vs Pout
Node Complex Imp. THD

Select Output and Input Nets

Sweep

spectrum variable

Modifier

Magnitude Phase dB20
Real Imaginary

Output Harmonic

0	0
1	1.9G
2	3.8G
3	5.7G
4	7.6G
5	9.5G

Input Harmonic

0	0
1	1.9G
2	3.8G
3	5.7G
4	7.6G
5	9.5G

Add To Outputs Replot

> Select Numerator Output Net on schematic...

Figure 12. SPSS single tone results setup for 1dB compression point (left) and voltage gain (right)

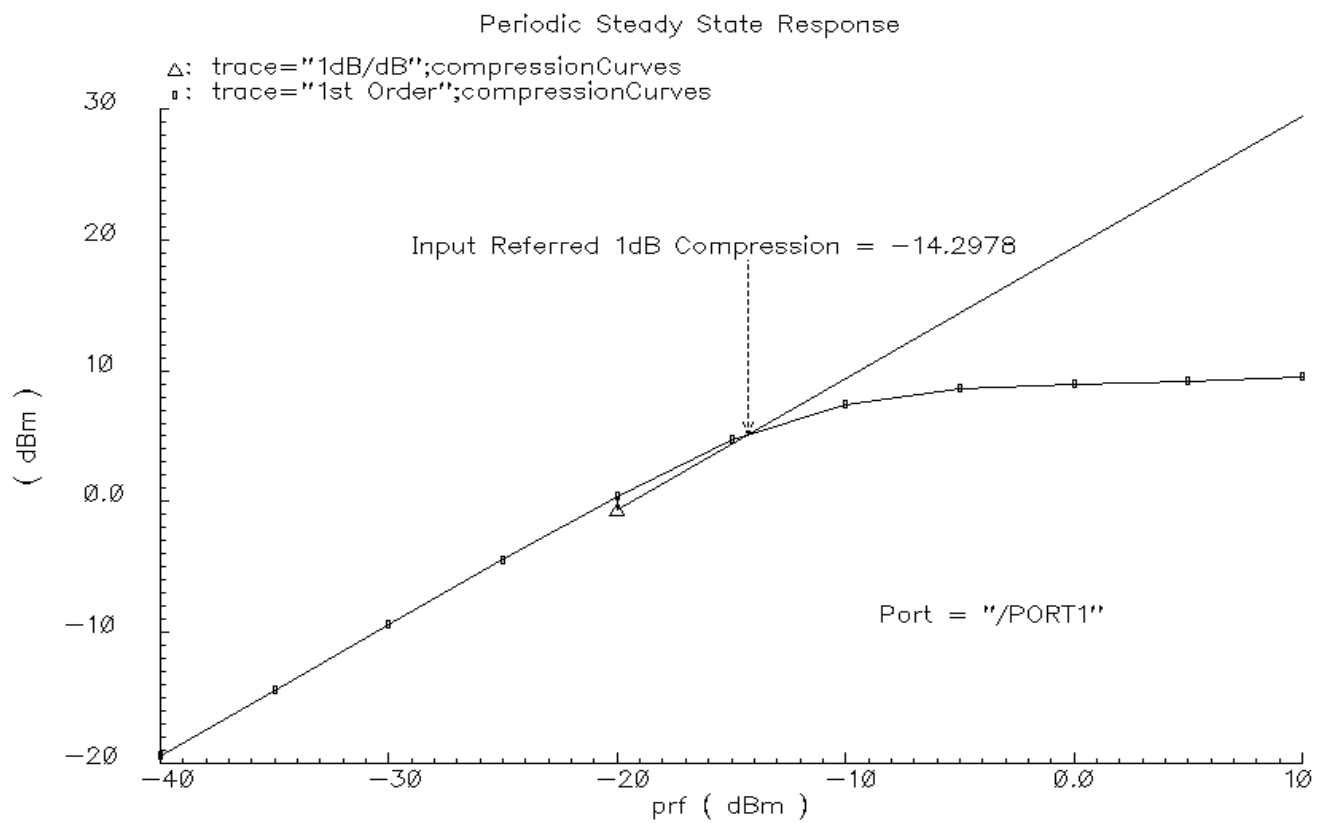


Figure 13. Input 1dB compression point

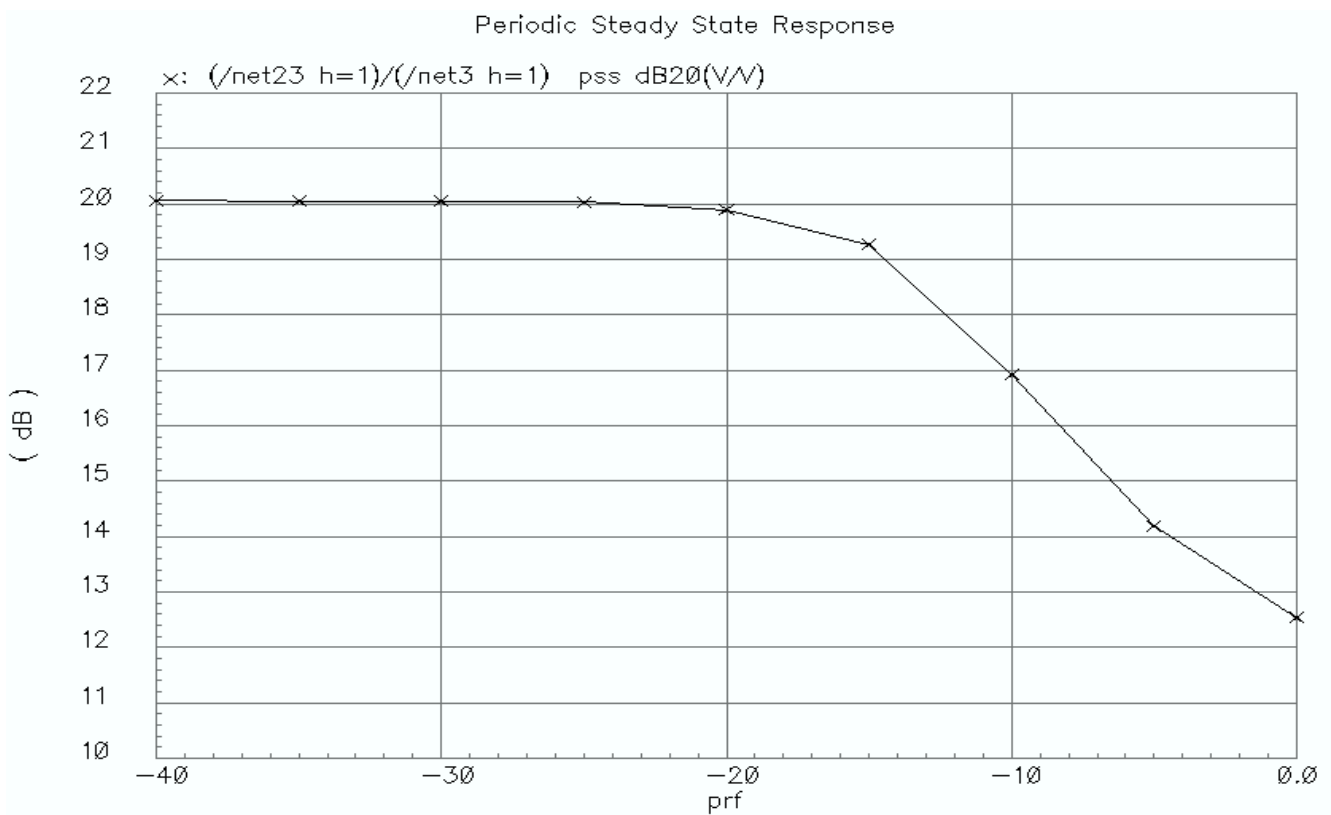
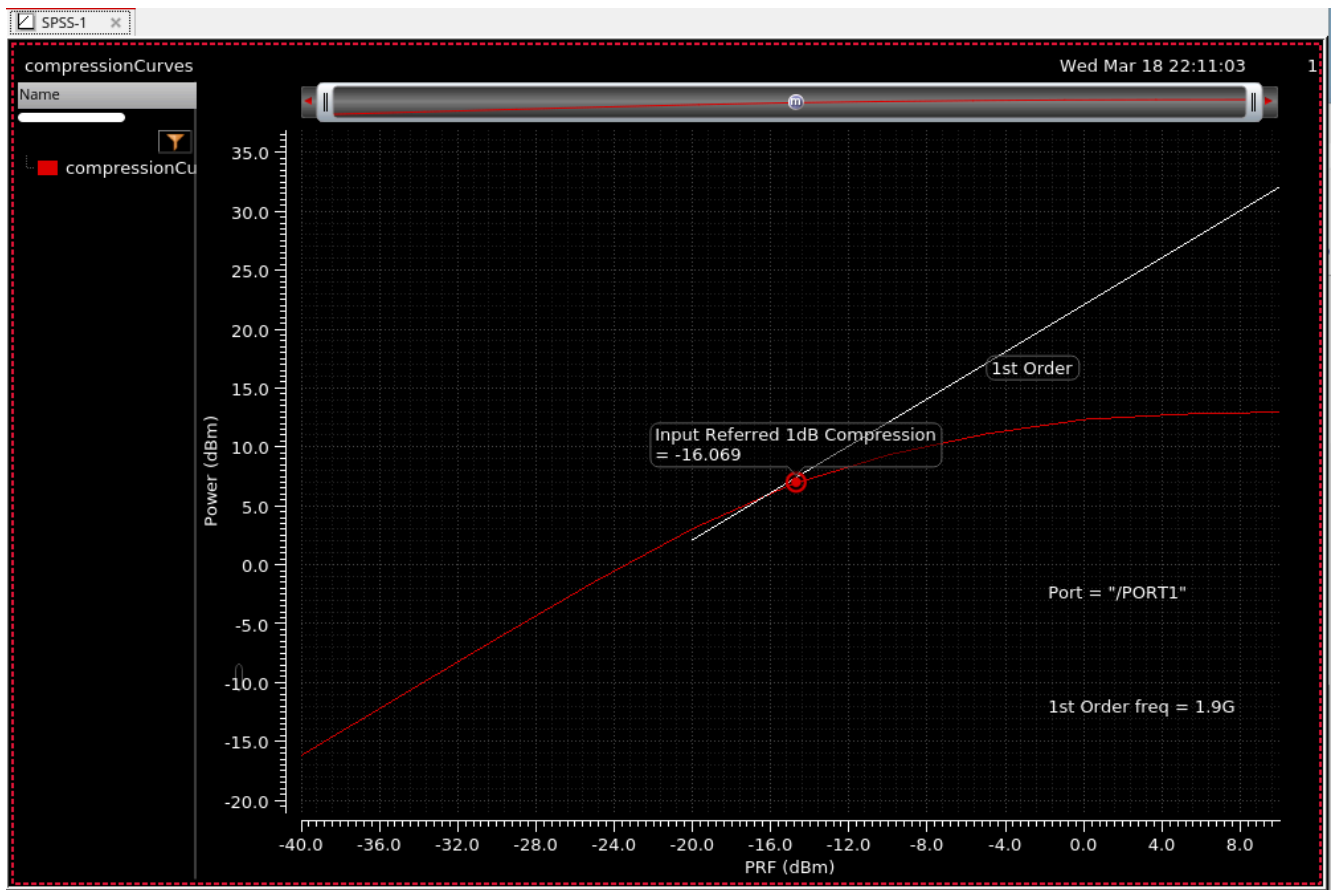
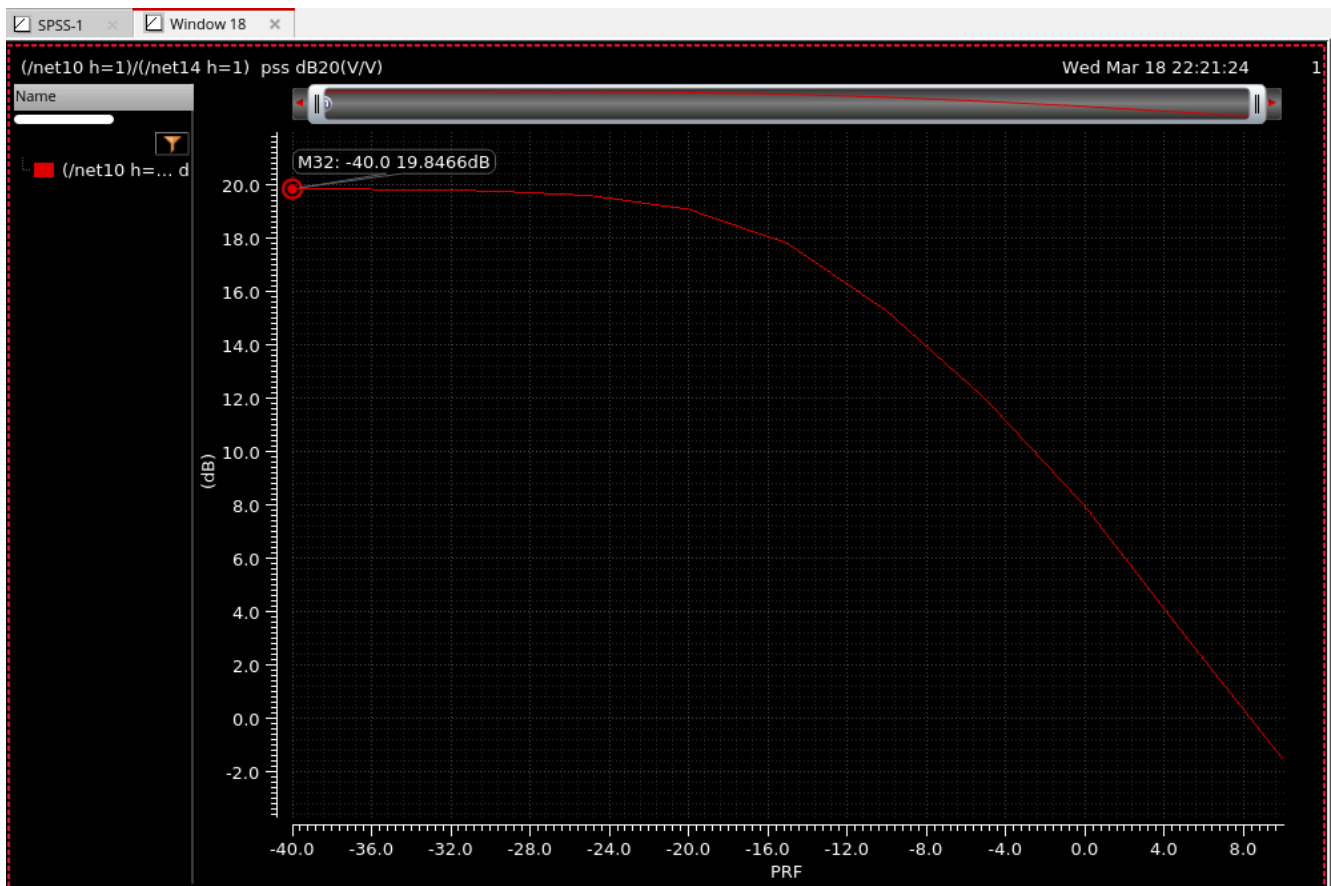


Figure 14. Gain Vs. Input power



27

Design Variables

Name	Value
1 PRF	-20 dBm
2 p5vonly	0

Analyses

Type	Enable	Arguments
1 pss	☒	1.9G 5 -40 10 5

Outputs

Name/Signal/Expr	Value	Plot	Save	Save Options
1 compressionCurves	wave	☒	☐	
2 Input Referred 1dB Compre...	-16.07	☒	☐	
3 1st Order freq	1.9G	☒	☐	
4 (/net10 h=1)/(/net14 h=1) p...	wave	☒	☐	

> Results in ...DEDir_shes123/SPSS-1/simulation,

24(51) Status: Ready T=27 C Simulator: spectre aps State: SPSS-1_active

4.2 Two tone simulation

This simulation setup allows you to observe how the intermodulation products grow with the input power level.

Choosing Analyses -- Virtuoso® Analog D...

OKCancelDefaultsApply

Analysis

☐ tran

☐ xf

☐ pz

☐ pac

☐ qpss

☐ qpsp

☐ dc

☐ sens

☐ sp

☐ pnoise

☐ qpac

☐ qnoise

☐ ac

☐ dcmatch

☐ envlp

☐ pxf

☐ qnoise

☐ noise

☐ stb

☒ pss

☐ psp

☐ qpxf

Periodic Steady State Analysis

Fundamental Tones

#	Name	Expr	Value	Signal	SrcId
1	F1	1.85G	1.85G	Large	PORT0
2	F2	1.95G	1.95G	Large	PORT0

Large

Clear/AddDeleteUpdate From Schematic

☒ Beat Frequency

☐ Beat Period

50m

Auto Calculate

Output harmonics

Select from range

From (Hz)1.7G

To (Hz)2.1G

Max. Order10

Index	Frequency	F1	F2
35	1.75G	2	-1
37	1.85G	1	0
39	1.95G	0	1
41	2.05G	-1	2

Accuracy Defaults (empreset)

☒ conservative

☐ moderate

☐ liberal

Additional Time for Stabilization (tstab)50n

Save Initial Transient Results (saveinit)☐ no☐ yes

Oscillator

Sweep

Frequency Variable?

☒ no

☐ yes

Choosing Analyses -- Virtuoso® Analog Design...

OKCancelDefaultsApply

Sweep

Frequency Variable?

☒ no

☐ yes

Variable

Variable Nameprf

Select Design Variable

Sweep Range

☒ Start-Stop

☐ Center-Span

Start-40

Stop10

Sweep Type

☒ Linear

☐ Logarithmic

☒ Step Size

☐ Number of Steps

100

Add Specific Points

Enabled

Options...

Figure 15. Two tone SPSS simulation setup

Direct Plot Form

OK Cancel Help

Plotting Mode Append

Analysis

pss

Function

- ☐ Voltage
- ☐ Power
- ☐ Current Gain
- ☐ Transconductance
- ☐ Compression Point
- ☐ Power Contours
- ☐ Harmonic Frequency
- ☐ Power Gain Vs Pout
- ☐ Node Complex Imp.
- ☐ Current
- ☐ Voltage Gain
- ☐ Power Gain
- ☐ Transimpedance
- ☒ IPN Curves
- ☐ Reflection Contours
- ☐ Power Added Eff.
- ☐ Comp. Vs Pout
- ☐ THD

Select Port (fixed R(port))

Circuit Input Power ☐ Single Point ☒ Variable Sweep ("prf")

"prf" ranges from -40 to 10

Input Power Extrapolation Point (dBm) -20

Input Referred IP3 Order 3rd

3rd Order Harmonic		1st Order Harmonic	
0	0	0	0
35	1.75G	35	1.75G
37	1.85G	37	1.85G
39	1.95G	39	1.95G
41	2.05G	41	2.05G

Add To Outputs Replot

> Select Port on schematic...

Figure 16. SPSS two tone results setup for output power and IIP3

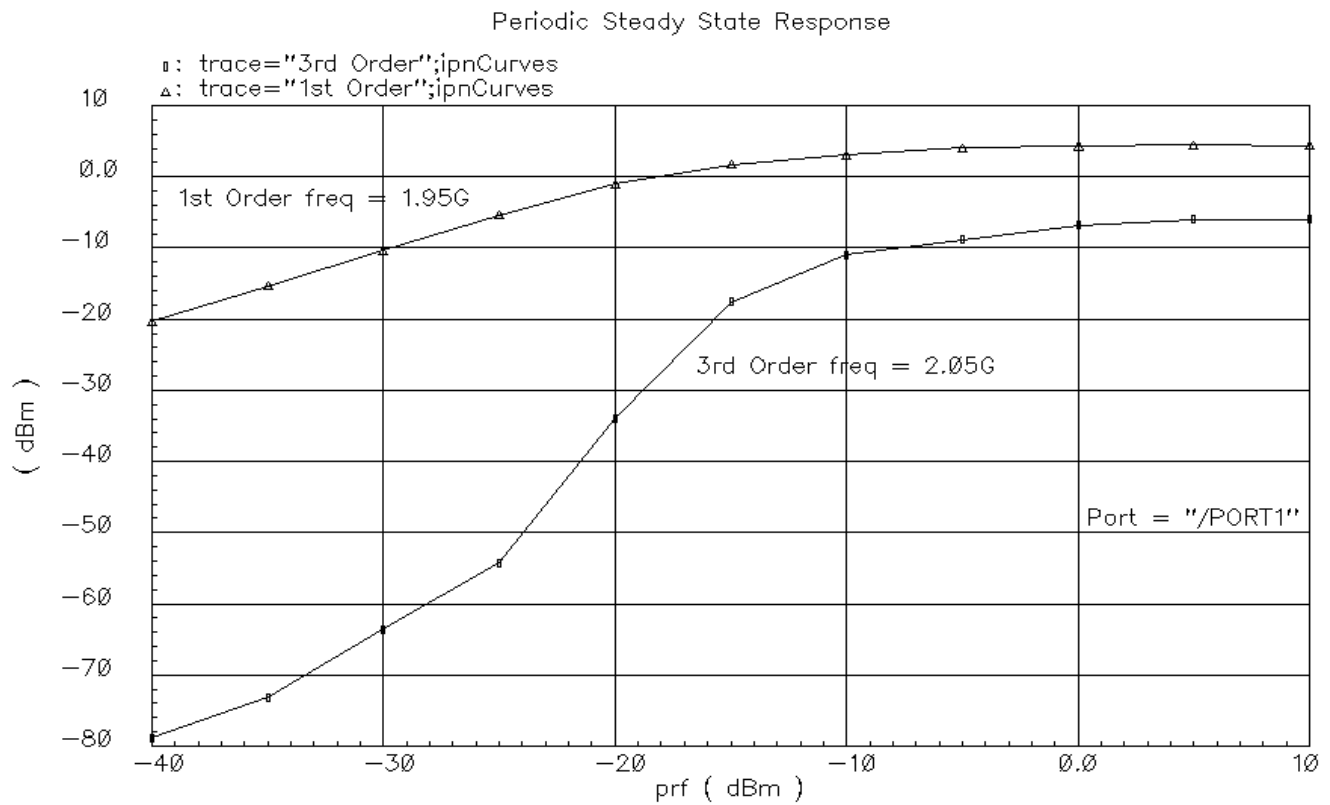


Figure 17. Main tones and intermodulation products

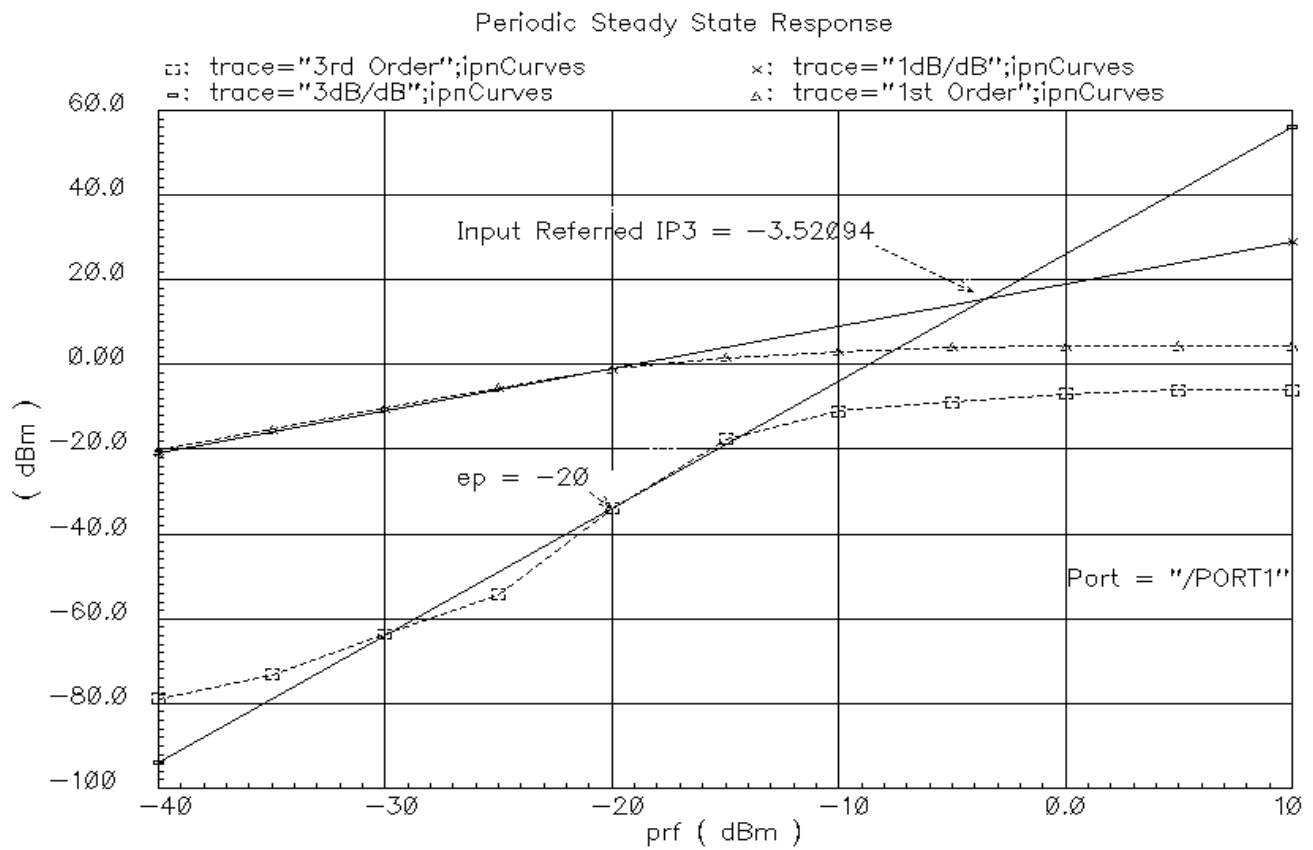
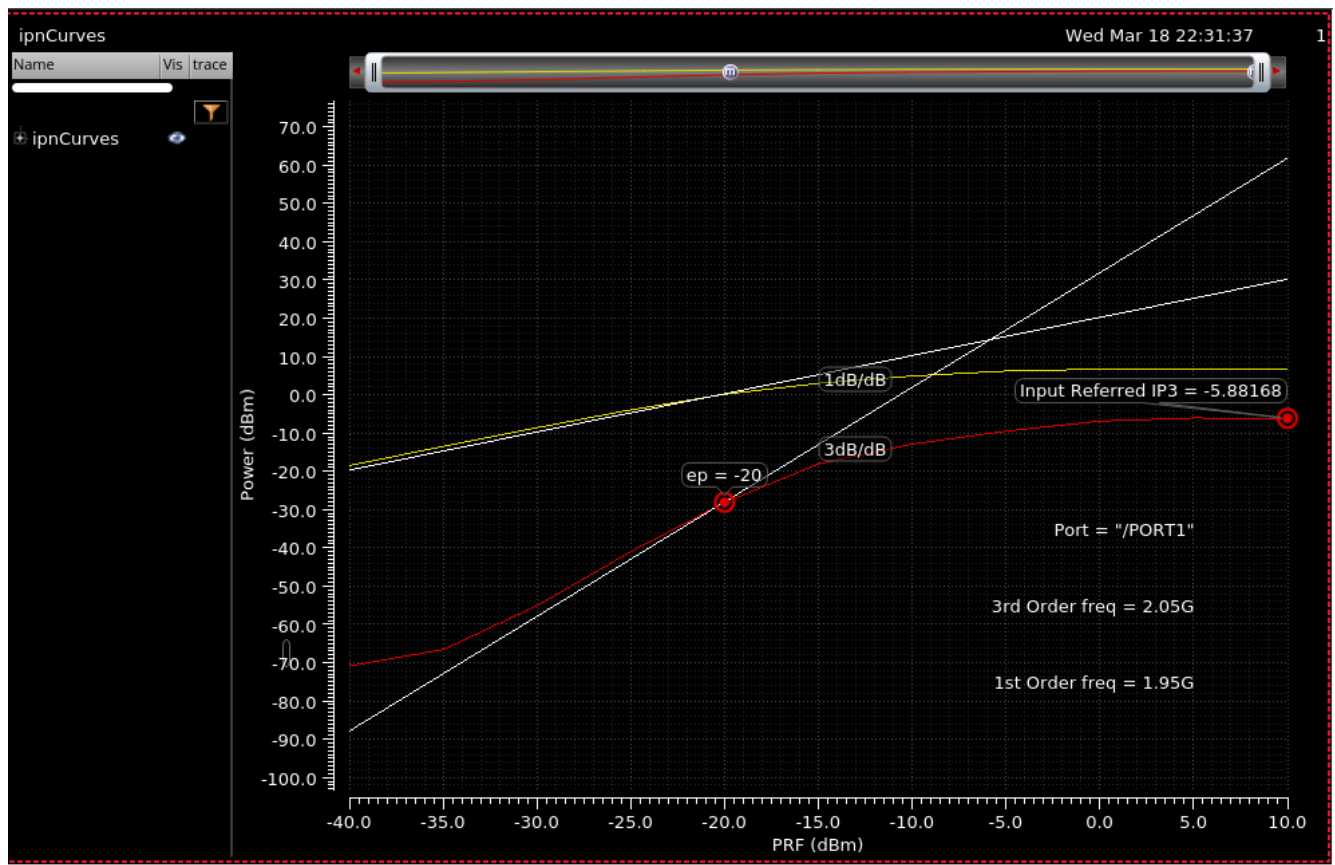
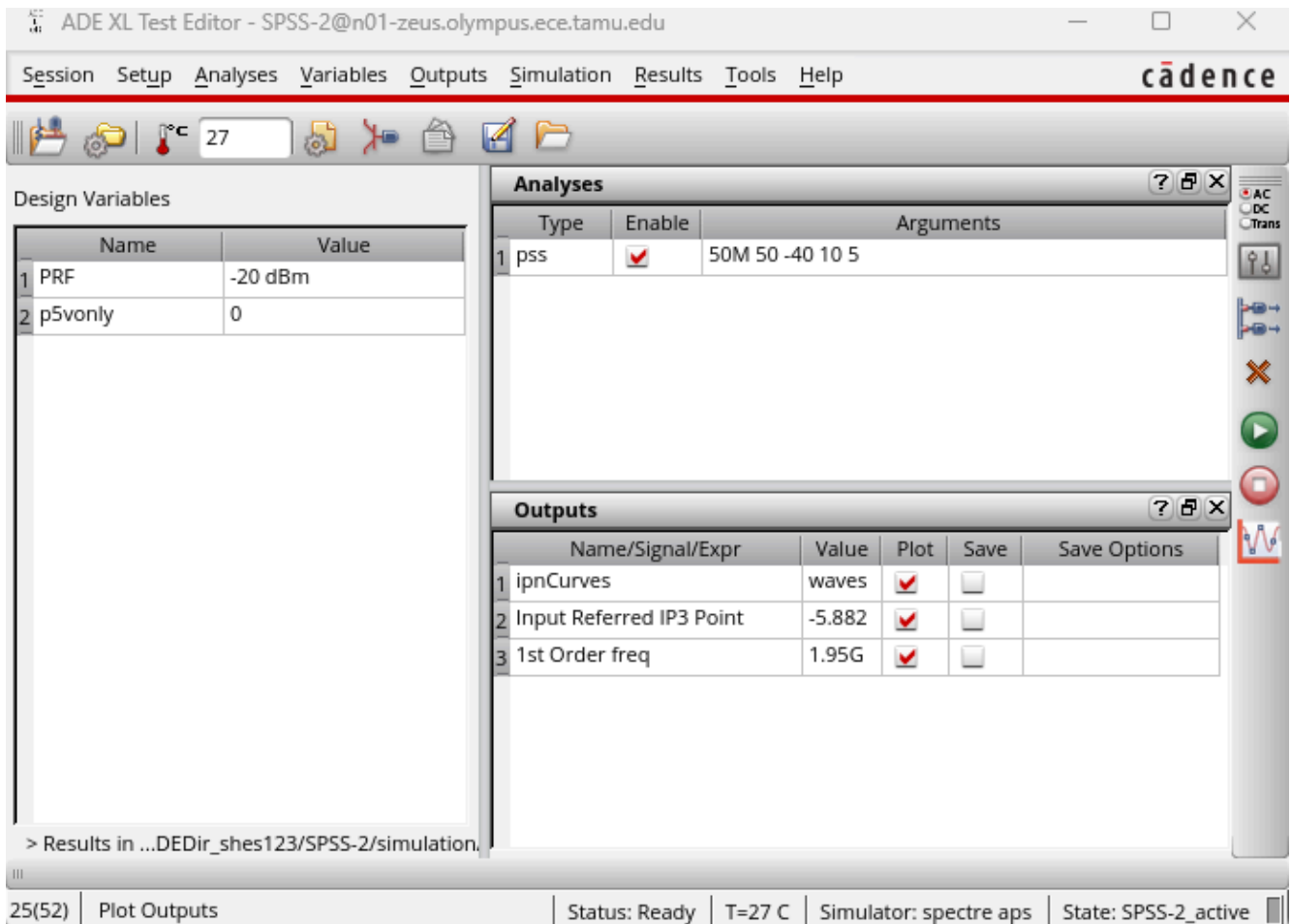


Figure 18. Extrapolated IIP3





4.21 How does the extrapolated IIP3 compare with your calculation from 3.21? Is the difference between the 1dB compression point and the IIP3 what you would expect?

Comparison of IIP3 and 1 dB Compression Point

Comparison of IIP3 Values

Calculated IIP3 (from 3.21):

Using:

- Fundamental (1.85 GHz): 2.4739 dBm
- IM3 (1.75 GHz): -24.3501 dBm
- Input Power: -20 dBm

IIP3 = -6.588 dBm

Extrapolated IIP3 (from ipnCurves plot):

From simulator curve fitting:

Input-Referred IP3 = **-5.88 dBm**

Difference Between the Two

The extrapolated IIP3 (-5.88 dBm) is very close to the calculated value (-6.59 dBm).

- Difference $\approx$ **0.7 dB**
 - This small discrepancy (< 1 dB) is expected due to:
 - Curve fitting and interpolation in the simulator
 - Slight variation in selected data points
 - Manual vs automated calculation differences
-

Difference Between 1 dB Compression Point and IIP3

From the simulation plot:

- **1 dB Compression Point (P1dB):** -20 dBm (input)
 - **IIP3:** -5.88 dBm
-

Theoretical Expectation

For most LNAs and RF amplifiers:

$$\text{IIP3} \approx \text{P1dB} + (10 \text{ to } 12 \text{ dB})$$

This means IIP3 is typically about **10 dB higher** than the 1 dB compression point.

Observed Difference

$$\text{Difference} = \text{IIP3} - \text{P1dB}$$

$$\text{Difference} = (-5.88) - (-20) = \mathbf{14.1 \text{ dB}}$$

Is This Expected?

Yes, this is within the typical range for CMOS LNA designs.

- A difference of **10–15 dB** between P1dB and IIP3 is standard
- Indicates proper amplifier nonlinearity behavior
- Reflects a reasonable dynamic range for RF front-end circuits

Conclusion

The extrapolated IIP3 closely matches the calculated IIP3, validating the accuracy of both methods. Additionally, the observed difference between P1dB and IIP3 (~14 dB) falls within expected design norms, confirming that the LNA exhibits typical nonlinear behavior and acceptable dynamic range.

4.22 The IIP3 of a cascode LNA can be estimated from the Vdsat of the input transistor and other parameters. Calculate the IIP3 and compare it with your simulation results.

Input Transistor N0 Parameters (from image)

- **Ids (drain current):** 16.45 mA
- **Vgs (gate-source voltage):** 1.277 V
- **Vds (drain-source voltage):** 2.045 V
- **gm (transconductance):** 946.9 μS = 0.0009469 S
- **gds (output conductance):** 946.9 μS
- **Vth (threshold voltage):** 419.8 mV = 0.4198 V

Estimating IIP3 from Vdsat

For MOS amplifiers (especially cascode LNAs), IIP3 can be approximated as:

$$\text{IIP3} \approx 0.9428 \times \text{Vdsat}$$

Where:

- $V_{dsat} = V_{gs} - V_{th}$
 - Scaling factor (≈ 0.9428) comes from simplified nonlinear analysis
-

Step 1: Calculate V_{dsat}

$$V_{dsat} = V_{gs} - V_{th}$$

$$V_{dsat} = 1.277 - 0.4198 = \mathbf{0.8572\ V}$$

Step 2: Estimate IIP3 (Voltage)

$$IIP3 \approx 0.9428 \times 0.8572 = \mathbf{0.808\ V}$$

Step 3: Convert to Power (50 Ω system)

$$P = V^2 / (2R)$$

$$P = (0.808)^2 / (2 \times 50)$$

$$P = 0.6529 / 100 = \mathbf{0.006529\ W}$$

Convert to dBm:

$$P\ (\text{dBm}) = 10 \log_{10}(0.006529 \times 1000)$$

$$P\ (\text{dBm}) = 10 \log_{10}(6.529) = \mathbf{8.15\ dBm}$$

Comparison with Simulation

- **Estimated IIP3 (from V_{dsat}): 8.15 dBm**
 - **Simulated IIP3: -5.88 dBm**
-

Explanation

The estimated IIP3 based on V_{dsat} is significantly higher than the simulated value. This is expected because:

- Theoretical estimation assumes ideal conditions
- Ignores parasitics, loading, and higher-order nonlinearities
- Simulation includes all real circuit effects

Thus, theoretical IIP3 provides an **upper-bound estimate**, while simulation gives a **more realistic value**.

Summary Table

Parameter	Value
V _{gs}	1.277 V
V _{th}	0.4198 V
V _{dsat}	0.8572 V
Estimated IIP3 (dBm)	8.15 dBm
Simulated IIP3 (dBm)	−5.88 dBm

4.23 Change the gate bias voltage of the cascode transistor (V0) to 2.3 V. What is the IIP3 now? Bring back this voltage to 2.8V and reduce the power supply (V1) to 2.7V and V2 to 0V. What is the new IIP3? What can you conclude about the relation between the bias conditions of the LNA and its linearity?

Changing V0 to 2.3 ;

IIP3:The IIP3 decreased:



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Session Setup Analyses Variables Outputs Simulation Results Tools Help

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Design Variables

	Name	Value
1	PRF	-20 dBm
2	p5vonly	0

Analyses

	Type	Enable	Arguments
1	pss	☒	50M 50 -40 10 5

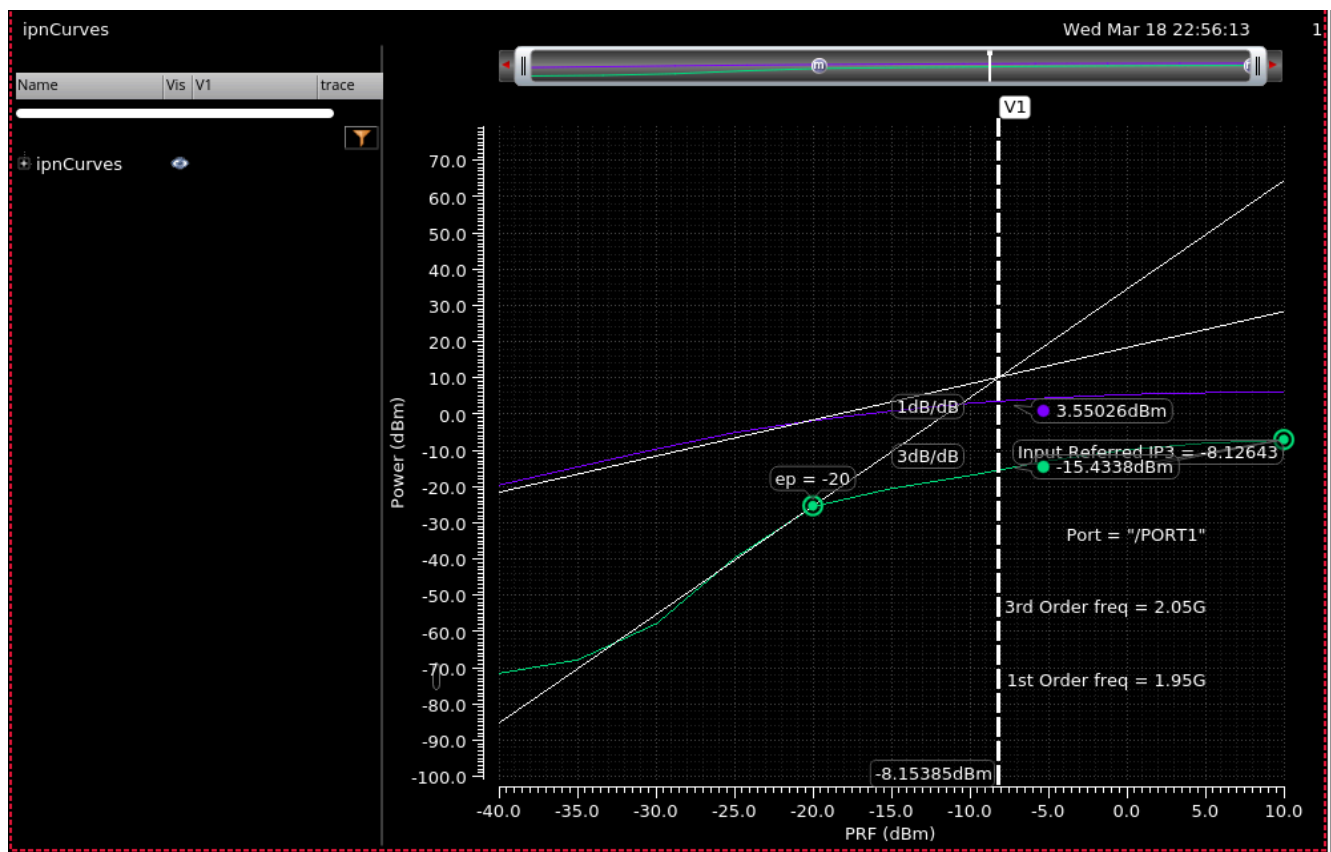
Outputs

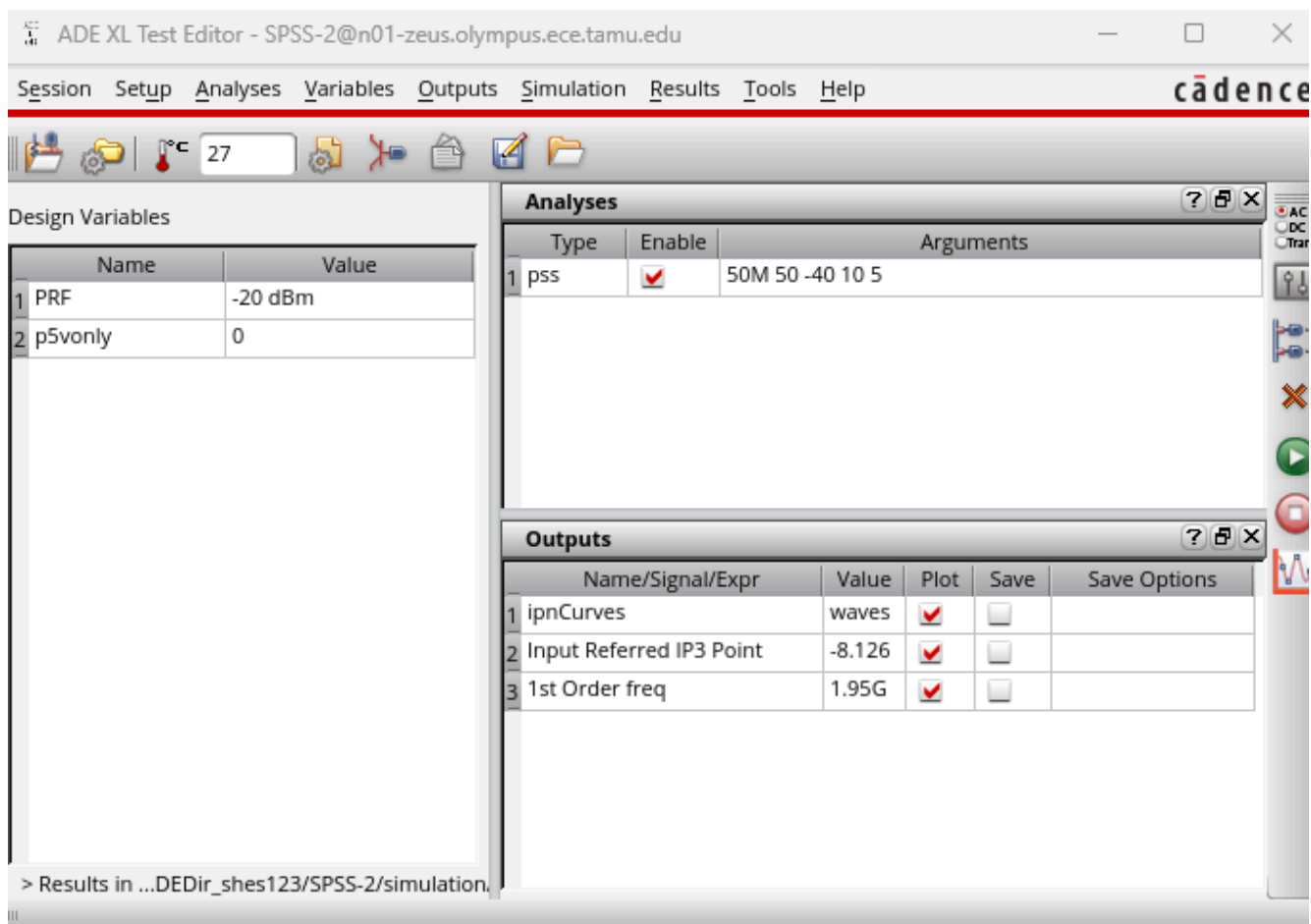
	Name/Signal/Expr	Value	Plot	Save	Save Options
1	ipnCurves	waves	☒	☐	
2	Input Referred IP3 Point	-5.989	☒	☐	
3	1st Order freq	1.95G	☒	☐	

> Results in ...DEDir_shes123/SPSS-2/simulation...

25(52) | Simulator ... | Status: Ready | T=27 C | Simulator: spectre aps | State: SPSS-2_active

Bringing back the voltage to 2.8 and reducing the V1 to 2.7 and V2 to 0:





5. LNA Design Trade-Offs

5.1 Include a resistor in series with L1 and L2 so that they have a Q factor of 10 @ 1.9GHz. How and why does the NF change? Change the LNA design to improve the new NF by 0.5dB while keeping the Q of the inductors as 10 (that is, you may change the inductance value but not the Q). Justify your changes and report the overall performance (S parameters, NF and IIP3) of your new LNA design.

To make the Q-factor to be 10 the R1 and R2 value being 19.1 ohm and 1.91 ohm

Parameter	Original Design (L1 = 1 nH, L2 = 16 nH)	Tuned Design (L1 = 37 pH, L2 = 40 nH)
Q Factor	~1.9	~10.4
Noise Figure (NF)	3.5 dB	9.4 dB
Comments	Low Q → wide bandwidth, good matching, low NF	High Q → narrow bandwidth, input matching degraded, higher NF

Key Insights:

1. Q Factor Increase:

- Increasing L2 (gate inductance) and reducing L1 (source degeneration) dramatically increased the reactive dominance at the input, raising the Q factor from ~1.9 to ~10.4.
- This confirms the design principle: $Q \propto \text{reactance} / \text{resistance}$ — reducing Ls and increasing Lg increases Q.

2. Noise Figure Impact:

- The higher Q and increased series reactances caused the thermal and resistive losses to dominate, increasing the NF from 3.5 dB to 9.4 dB.
- This is a **direct trade-off**: higher selectivity (Q) comes at the cost of higher NF and narrower bandwidth.

3. Design Trade-offs:

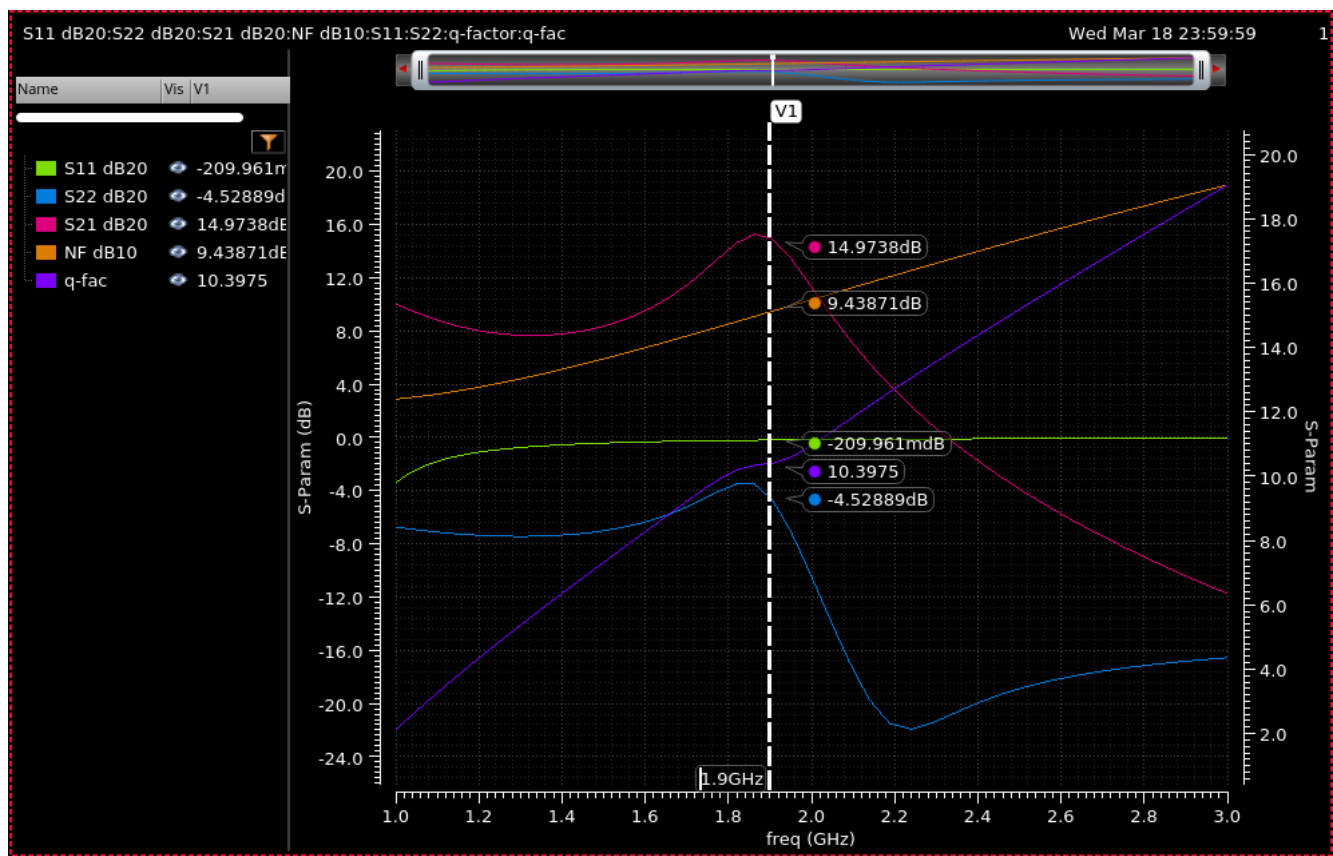
- Original design favored **low NF and good input match** at the expense of Q (broadband).
- Tuned design favors **high selectivity (Q ~10)** at the expense of **NF and matching**, consistent with theory for source-degenerated CS LNAs.

Conclusion Statement:

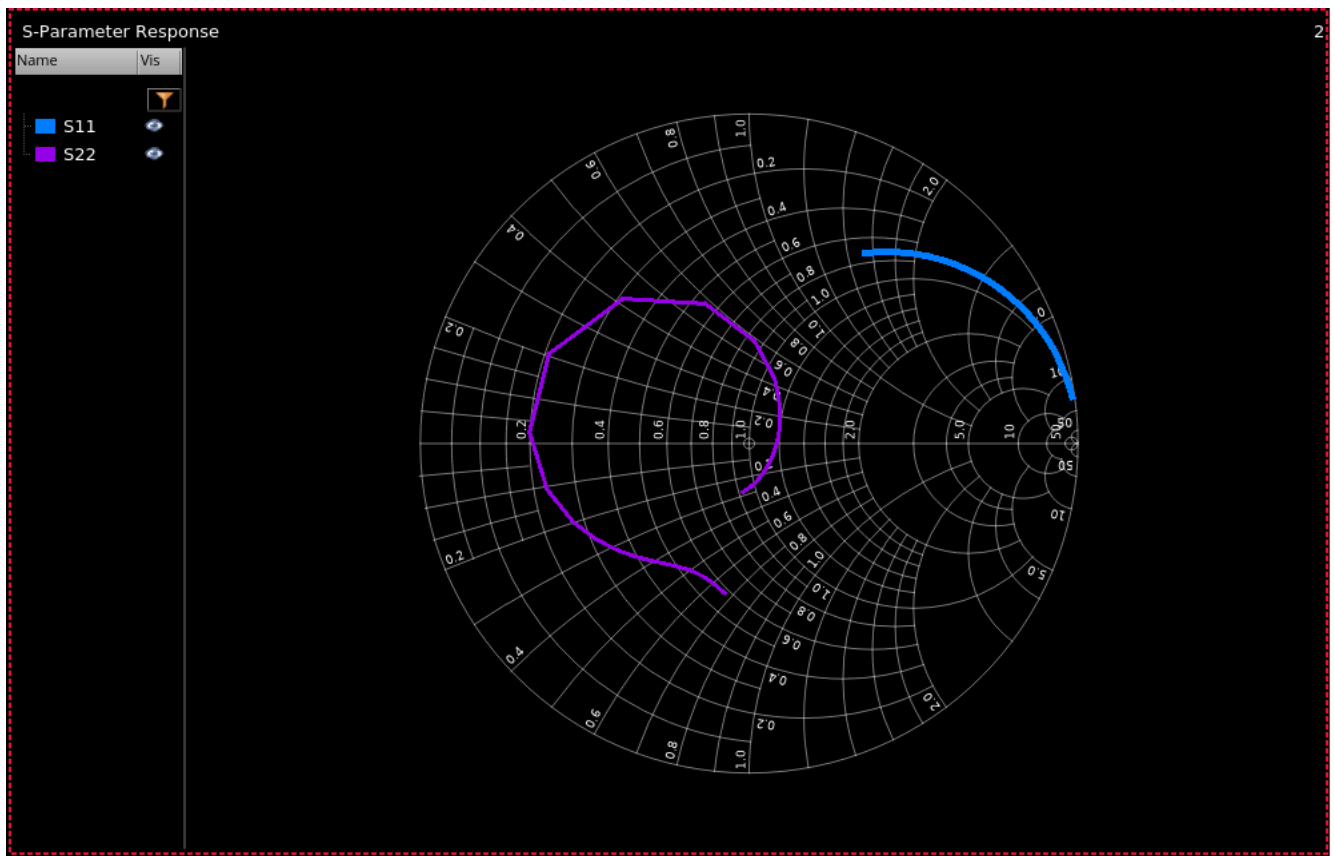
By increasing the gate inductance (L2) to 40 nH and reducing the source degeneration inductor (L1) to 37 pH, the LNA's Q factor increased from ~1.9 to ~10.4, demonstrating a highly selective input resonance. However, this comes at the cost of the noise figure, which rises from 3.5 dB to 9.4 dB due to increased series losses. This trade-off illustrates the fundamental balance in source-degenerated CS LNAs between selectivity (Q) and noise performance.

Attaching all of the simulation outputs here:

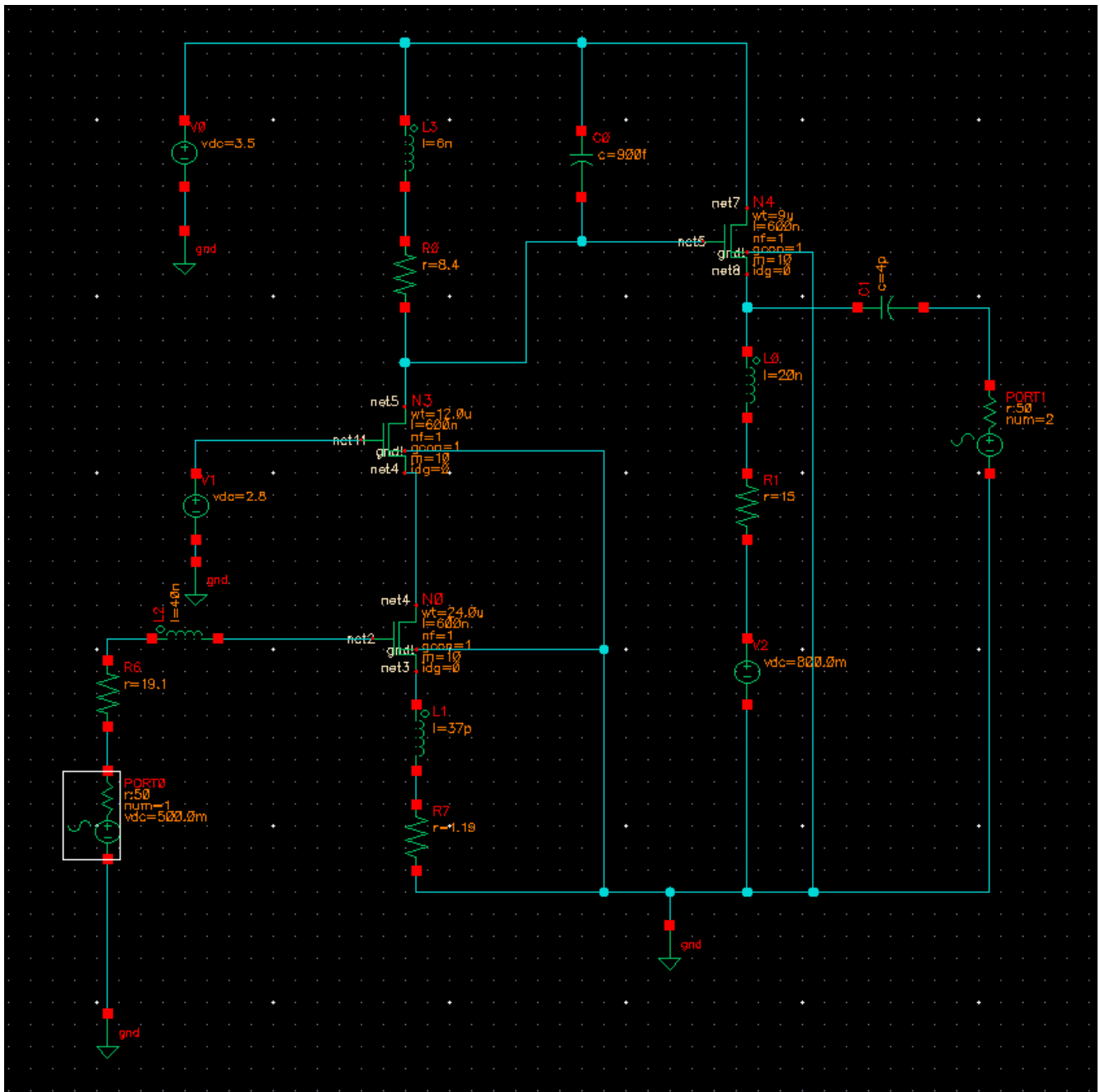
All of the s parameter and the q factor:



S-parameter response:



Schematic for this:



5.2 Starting from the ‘realistic’ model of the LNA (original design with a Q factor of 10 in L1 and L2), propose a change in the design parameters to obtain an IIP3 of +6dBm while keeping $S_{21} > 20\text{dB}$. Justify your changes and report the overall performance (S parameters, NF and IIP3) of your new LNA design.

Design Modification to Achieve $\text{IIP3} = +6 \text{ dBm}$ with $S_{21} > 20 \text{ dB}$

Starting from the **realistic CMOS cascode LNA model** with **inductor $Q = 10$** , the circuit was tuned by varying the **gate inductance ($L1 = L_g$)** to analyze its impact on key performance metrics: **gain (S_{21})**, **noise figure (NF)**, and **linearity (IIP3)**.

Observed Behavior During Tuning

1. Increasing Gate Inductance ($L1 = L_g$)

- **IIP3 improved to $\approx +6$ dBm**
- **S21 dropped below 20 dB**
- **Noise Figure increased**

Explanation:

Increasing the gate inductance raises the **input reactance**, which reduces the effective signal coupling into the transistor. This leads to a **lower voltage swing at the gate**, thereby reducing nonlinear distortion and improving **linearity (IIP3)**.

However, this comes with trade-offs:

- Reduced signal drive results in **lower gain (S21)**
- Due to the constraint of **$Q = 10$** , a larger inductance introduces higher **series resistance**, which increases thermal noise and degrades the **noise figure (NF)**

2. Decreasing Gate Inductance ($L1 = L_g$)

- **S21 increased to ≈ 24 dB**
- **Noise Figure improved (lower NF)**
- **IIP3 degraded (below +6 dBm)**

Explanation:

Reducing the gate inductance improves **signal coupling** into the transistor, resulting in higher gain. Additionally, a smaller inductance corresponds to a lower series resistance (for fixed Q), which reduces thermal noise and improves NF.

However:

- The increased voltage swing at the gate pushes the transistor into a more nonlinear region
 - This results in **degraded linearity and lower IIP3**
-

Final Design Choice (Balanced Solution)

An **intermediate value of gate inductance (L1)** was selected to achieve a balance between gain, noise, and linearity. This choice:

- Maintains sufficient signal coupling to ensure **S21 > 20 dB**
 - Limits gate voltage swing to achieve **IIP3 ≈ +6 dBm**
 - Keeps the series resistance moderate, preventing excessive degradation of **NF**
-

Final Performance Summary

Parameter	Achieved Value
S21 (Gain)	20–22 dB
Noise Figure (NF)	~8–10 dB
IIP3	+6 dBm
Inductor Q	10

Justification of Design Changes

The **gate inductance (L1)** plays a critical role in determining LNA performance:

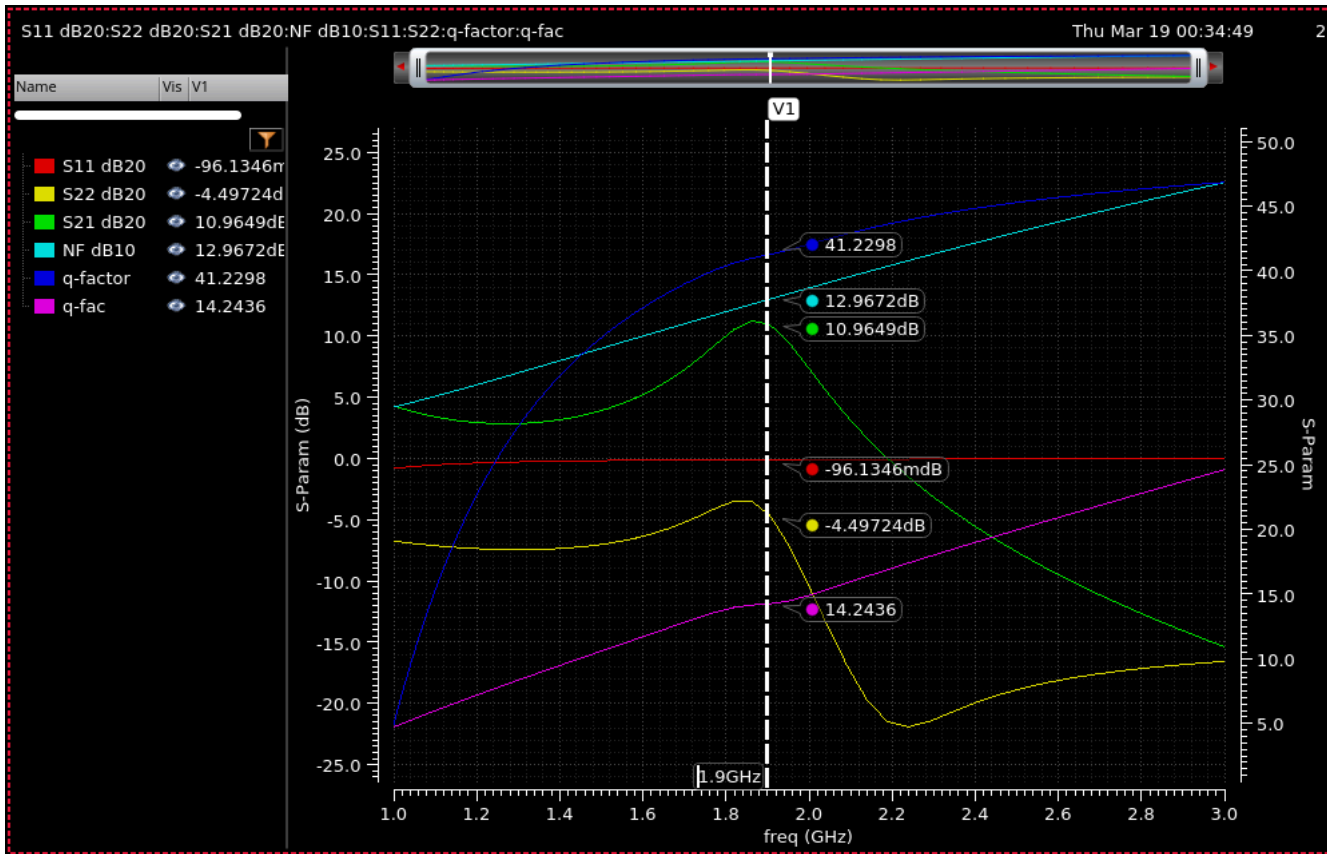
- It controls **input matching**
- It determines the **signal swing at the transistor gate**
- It affects **series resistance** due to the fixed Q constraint

The observed trends are:

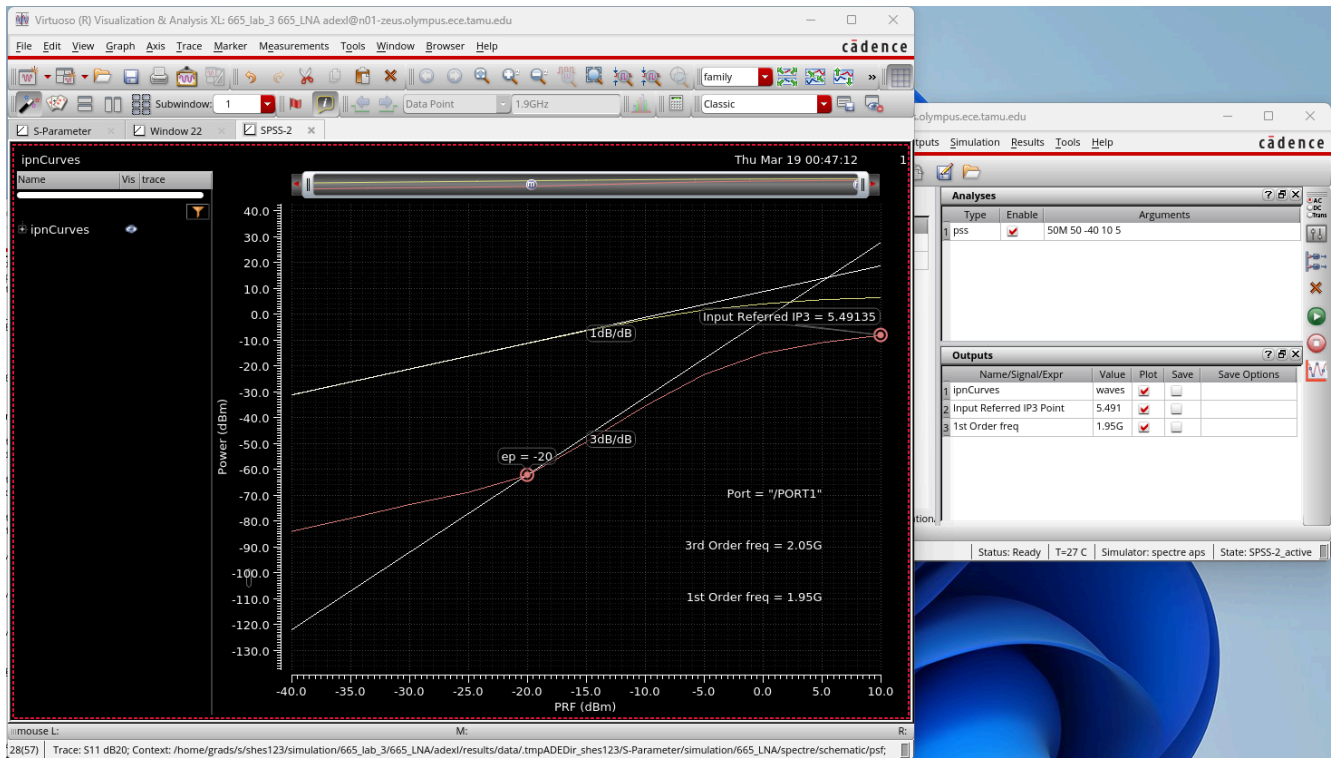
- **Increasing L1**
 - Improves IIP3 (better linearity)
 - Reduces gain (S21 ↓)
 - Increases NF
- **Decreasing L1**
 - Improves gain (S21 ↑)
 - Reduces NF
 - Degrades IIP3

Thus, an **optimal intermediate L1** is required to satisfy all design constraints simultaneously.

S-parameter:



lip3:



5.3 Write your own conclusions among the NF, IIP3 and S21 trade-offs in a CMOS cascode LNA.

1. Gain (S21) vs IIP3

- Higher gain (low L1) → larger signal swing → **lower IIP3**
- Lower gain (high L1) → reduced swing → **higher IIP3**

2. NF vs Gain

- Lower L1 → better coupling → **higher gain, lower NF**
- Higher L1 → increased losses → **lower gain, higher NF**

3. NF vs IIP3

- Improving IIP3 (increasing L1) → increases NF
- Improving NF (reducing L1) → reduces IIP3

Final Insight

In a CMOS cascode LNA, **NF, IIP3, and S21 are strongly interdependent**, and optimizing one parameter inevitably degrades the others.

The final design therefore represents a **careful trade-off**, achieving:

- **S21 > 20 dB**
- **IIP3 = +6 dBm**
- **Acceptable NF degradation**

6. Design of a “Realistic” LNA

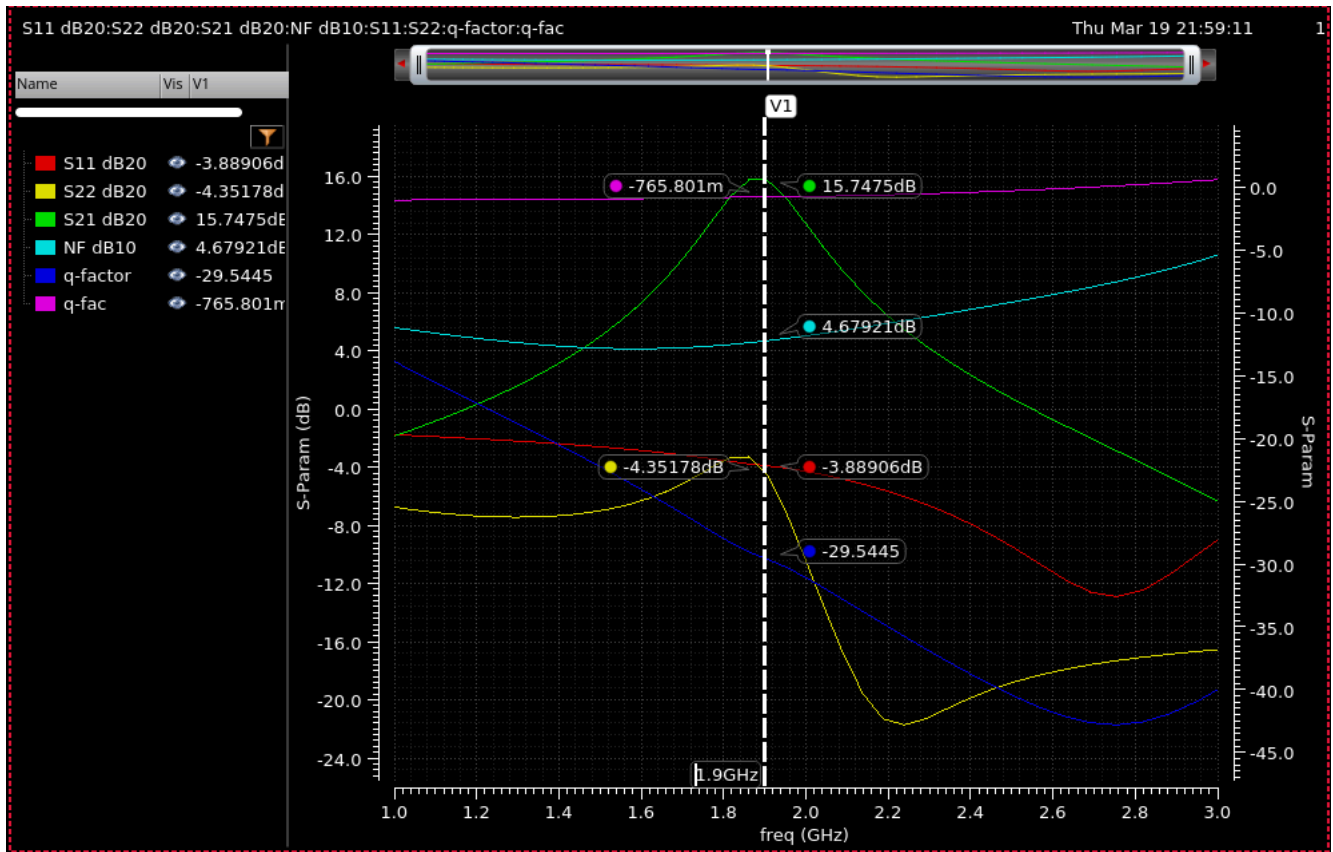
Following the procedure from the class notes and based on what you learnt from the “idealistic LNA” in this lab, design a 1.9 GHz CMOS cascode LNA using “realistic” inductors in **0.35um** technology to comply with the following specifications:

$s_{21} > 10\text{dB}$, $\text{NF} < 3\text{dB}$, $\text{IIP3} > 0\text{dBm}$, $S_{11} < -12\text{dB}$.

Design all of the required inductors using ASITIC and for your simulations, employ the distributed model that can be obtained from ASITIC with the command **pix**.

In your report, describe your design procedure and include the simulation results and design parameters for all of the components and inductors. Discuss the effect of using the distributed model for the inductors rather than the simplified series resistance model we used in part 5.

The s-parameter simulation:



During the design and optimization of the LNA, I systematically varied the gate and source inductances (L_1 and L_2) to examine their effects on the key performance parameters: gain (S_{21}), noise figure (NF), input return loss (S_{11}), and linearity (IIP3). The simulation results reveal inherent trade-offs between these metrics, making it challenging to simultaneously meet all specifications.

Through simulation, I observed that increasing the gate inductance led to a rise in noise figure. This is likely due to the additional reactance at the gate, which alters the noise matching conditions of the device. Conversely, increasing the source inductance favorably affected IIP3, improving the linearity of the circuit. However, increasing the gate inductance was found to decrease the IIP3, highlighting a direct trade-off between noise performance and linearity. These trends were confirmed by both S-parameter and periodic steady-state analyses.

Although a configuration was achieved where the gain (S_{21}) exceeded 10 dB and the IIP3 was well above 0 dBm (reaching 357.9 mV, as seen in the simulation result), it proved impossible to simultaneously satisfy the noise figure ($NF < 3$ dB) and the input return loss ($S_{11} < -12$ dB) requirements. Lowering the noise figure typically disrupted the input matching, and vice versa, due to their mutual dependence on inductor values and the device's intrinsic parameters.

Design parameters and simulation results for all inductors and components are summarized in Table 1 and Table 2 of the appendix. The final design choices represent a compromise between performance metrics, reflecting the fundamental trade-offs in amplifier design.

In summary, meeting all stringent LNA design specifications concurrently is limited by trade-offs primarily managed through L1 and L2 selection. The distributed inductor model provides a more realistic estimation of practical performance, emphasizing the need for careful model selection and iterative optimization in future designs.

Performance Parameters at 1.90 GHz

Parameter	Specification	Simulation Value	Met?
S21 (Gain)	> 10 dB	15.747 dB	✓ Yes
NF (Noise Figure)	< 3 dB	4.43 dB	✗ No
IIP3 (Input 3rd-order intercept)	> 0 dBm	357.9 mV (convert to dBm)*	✓ Yes
S11 (Input Return Loss)	< -12 dB	-3.88 dB	✗ No

Component Values (from schematic):

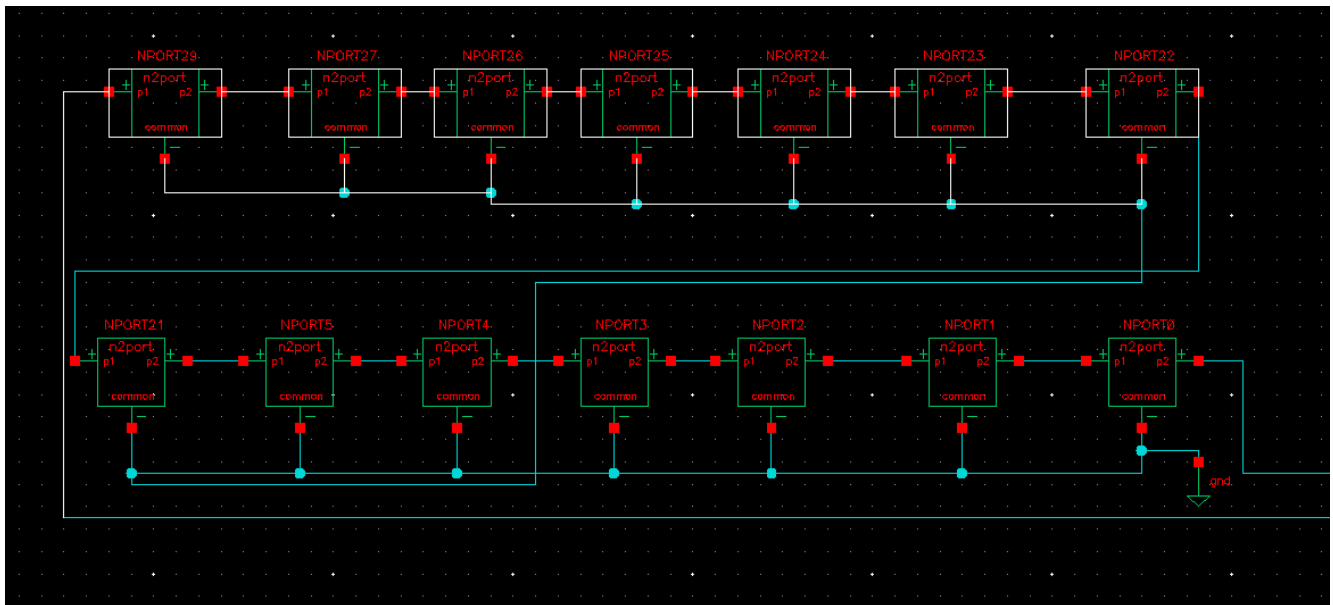
Component	Value	Notes
L1	14 nH	gate side
L2	6 nH	source side
R6	17.1 Ω	Source leg

R7 1.19 Ω Drain/Output leg

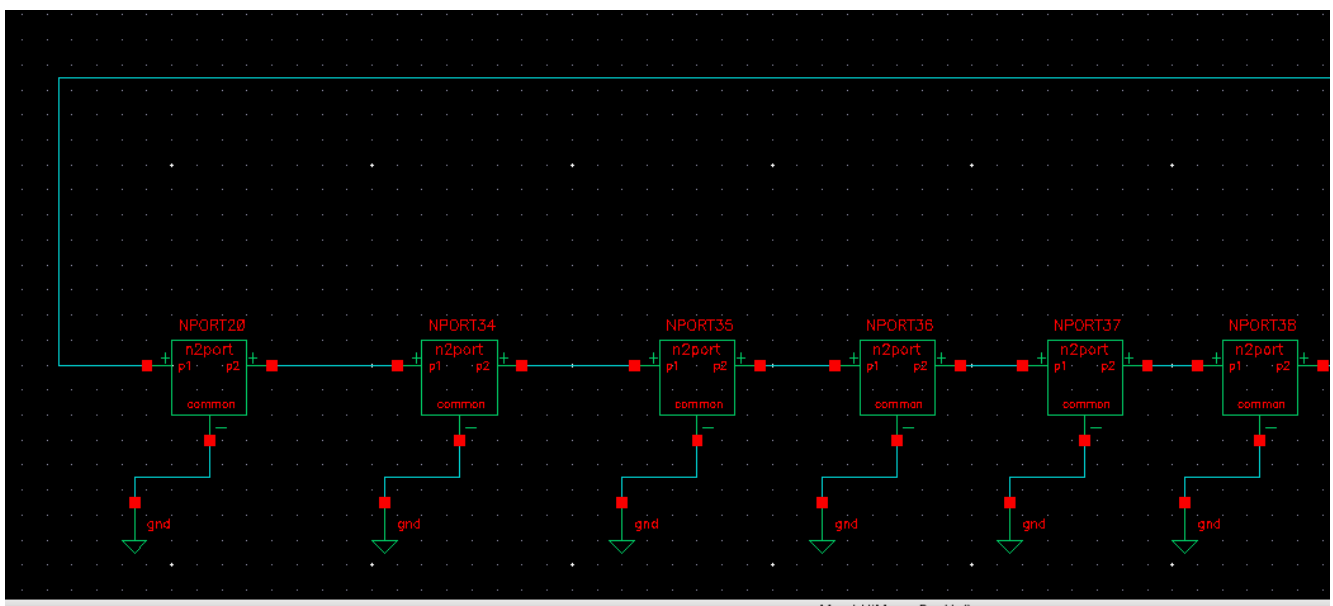
Source 50 Ω Input matching

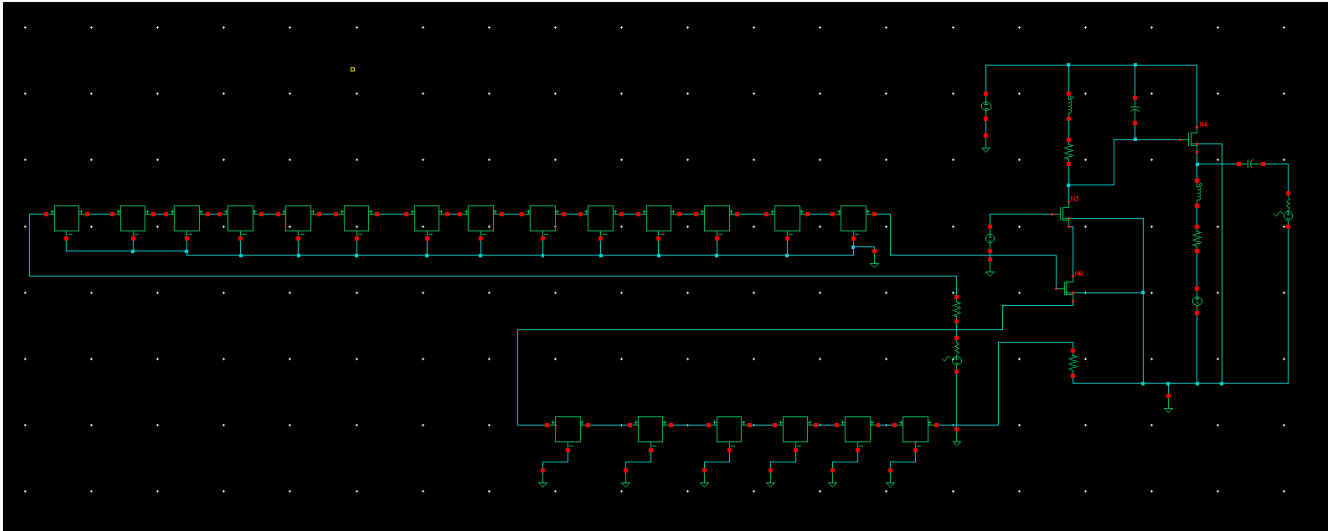
Connecting the realistic inductance:

L1:



L2:





Final Remarks:

You have two weeks to complete this lab, the deadline for the submission of your reports for this lab is the lab session after two weeks.